MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE BME

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic monostable multivibrator module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1065 failures per million operating hours for class I or 0.5180 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BME.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/257-1 for class I and M28787/257-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3.0 to 18.0 Vdc ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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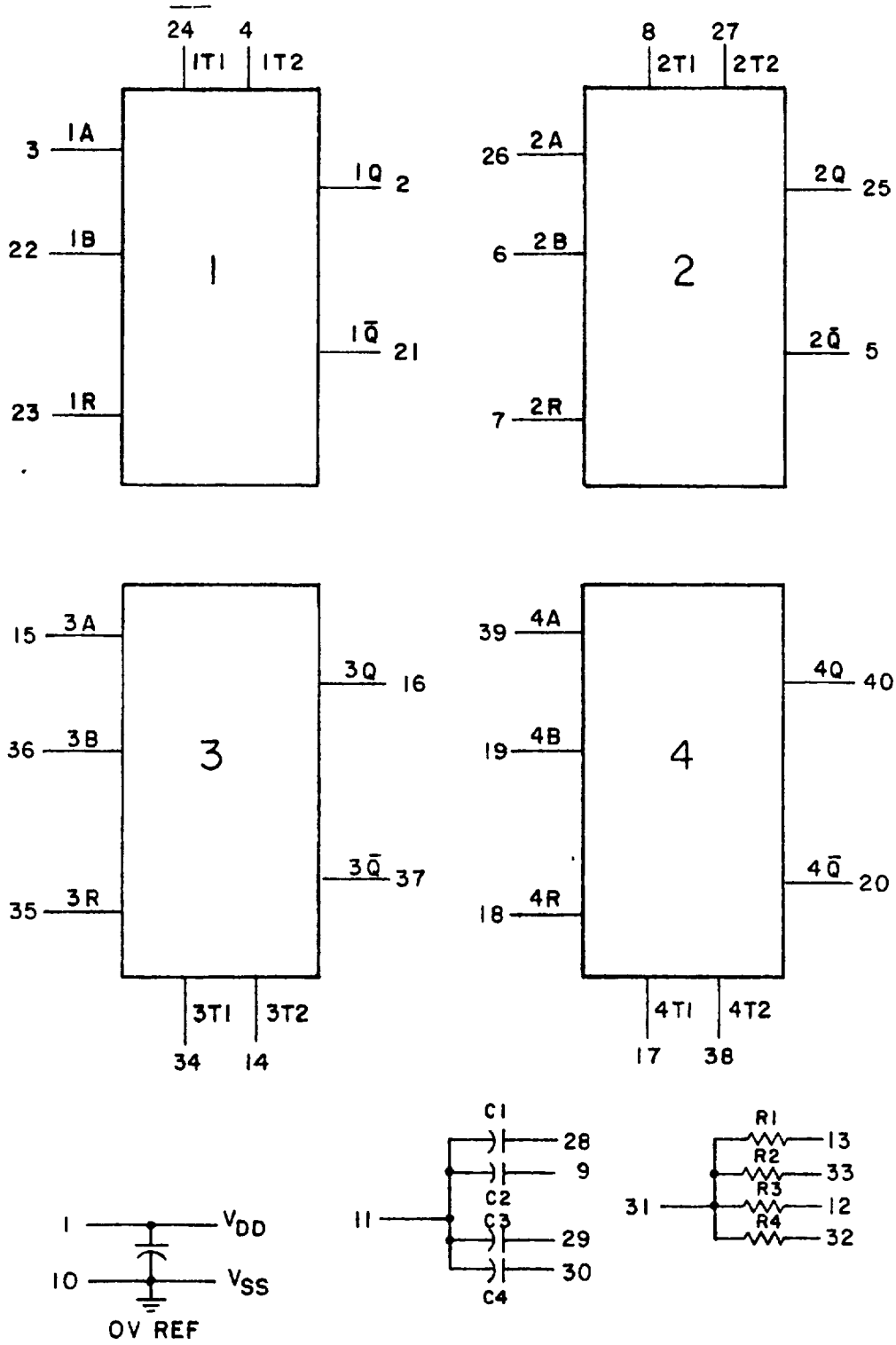


FIGURE 1. BME module function diagram, monostable multivibrator.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation Initial End-of-life	III III	S S	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	MΩ MΩ
B1-B16	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C16	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D8	Output voltage, low level (I_{OL} = 0)	VI	X	0	50	0	50	0	50	0	50	0	50	mV
E1-E8	Output voltage, high level (I_{OH} = 0)	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
F1-F8	Output voltage, low level (V_{DD} = 5 V)	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G8	Output voltage, high level (V_{DD} = 5 V)	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
H1-H8	Output voltage, low level (V_{DD} = 15 V)	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
J1-J8	Output voltage, high level (V_{DD} = 15 V)	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
K1-K16	Input current, low level	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
L1-L16	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
M1	Power supply current, low level	XIV	X	0	40	0	40	0	40	0	40	0	40	µA
M2	Power supply current, high level	XIV	X	0	40	0	40	0	40	0	40	0	40	µA
N1, N2	Output pulse width, match, V_{DD} = 5 V	XV	S	-400	400	-386	386	-416	416	-352	352	-432	432	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N3, N4	Output pulse width, match, V_{DD} = 5 V	XV	S	-55	55	-53	53	-57	57	-48	48	-59	59	µs
P1, P2	Output pulse width, match, V_{DD} = 10 V	XVI	Q	-500	500	-482	482	-520	520	-440	440	-540	540	ns
P3, P4	Output pulse width, match, V_{DD} = 10 V	XVI	Q	-110	110	-106	106	-114	114	-97	97	-119	119	µs
Q1, Q2	Output pulse width, match, V_{DD} = 15 V	XVII	Q	-500	500	-482	482	-520	520	-440	440	-540	540	ns
Q3, Q4	Output pulse width, match, V_{DD} = 15 V	XVII	Q	-130	130	-126	126	-135	135	-114	114	-140	140	µs
R1-R4	Output pulse width check, V_{DD} = 5 V	XVIII	S	1.65	3.35	1.59	3.23	1.72	3.48	1.45	2.95	1.78	3.62	µs
R5-R8	Output pulse width check, V_{DD} = 5 V	XVIII	S	4.65	9.35	4.49	9.02	4.84	9.72	4.08	8.23	5.02	10.1	µs
R9-R12	Output pulse width check, V_{DD} = 5 V	XVIII	S	23.3	46.7	22.5	45.1	24.2	48.6	20.5	41.1	25.2	50.4	µs
R13-R16	Output pulse width check, V_{DD} = 5 V	XVIII	S	235	470	227	454	244	489	207	414	254	508	µs
S1-S4	Output pulse width check, V_{DD} = 10 V	XIX	Q	1.25	3.75	1.21	3.63	1.30	3.90	1.10	3.30	1.35	4.05	µs
S5-S8	Output pulse width check, V_{DD} = 10 V	XIX	Q	3.75	11.3	3.62	10.9	3.90	11.8	3.30	9.94	4.05	12.20	µs
S9-S12	Output pulse width check, V_{DD} = 10 V	XIX	Q	25	75	24	72	26	78	22	66	27	81	µs
S13-S16	Output pulse width check, V_{DD} = 10 V	XIX	Q	275	825	265	796	286	858	242	726	297	891	µs
T1-T4	Output pulse width check, V_{DD} = 15 V	XX	Q	1.25	3.75	1.21	3.62	1.30	3.90	1.10	3.30	1.35	4.05	µs

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T5-T8	Output pulse width check, V_{DD} = 15 V	XX	Q	4	12	3.9	11.6	4.2	12.5	3.5	10.6	4.3	13.0	µs
T9-T12	Output pulse width check, V_{DD} = 15 V	XX	Q	30	90	29	87	31	94	26	79	32	97	µs
T13-T16	Output pulse width check, V_{DD} = 15 V	XX	Q	325	975	314	941	338	1014	286	858	351	1053	µs
U1-U8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XXI	S	35	350	35	350	37	375	29	292	40	404	ns
U9-U16	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XXI	Q	83	830	83	830	90	906	65	657	99	992	ns
V1-V8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XXII	S	26	260	26	260	27	273	23	231	28	287	ns
V9-V16	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XXII	Q	74	740	74	740	77	778	65	654	82	821	ns
W1-W8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XXIII	Q	15	150	15	150	16	161	12	126	17	173	ns
W9-W16	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XXIII	Q	13	130	13	130	13	135	11	118	14	141	ns
X1-X8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XXIV	Q	11	110	11	110	11	118	9	92	12	127	ns
X9-X16	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXIV	Q	10	100	10	100	10	104	9	90	10	109	ns
Y1-Y8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXV	S	66	660	66	660	67	671	63	636	68	682	ns

See footnotes at end of table.

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						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y9-Y16	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXV	S	110	1100	110	1100	111	1110	107	1076	112	1122	ns
Y17-Y24	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXV	Q	86	860	86	860	89	892	78	788	92	928	ns
Y25-Y32	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXV	Q	130	1300	130	1300	133	1335	122	1228	136	1368	ns
Z1-Z8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXVI	S	66	660	66	660	67	671	63	636	68	682	ns
Z9-Z16	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXVI	S	110	1100	110	1100	111	1110	107	1076	112	1122	ns
Z17-Z24	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXVI	Q	86	860	86	860	89	892	78	788	92	928	ns
Z25-Z32	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXVI	Q	130	1300	130	1300	133	1332	122	1228	136	1368	ns
AA1-AA8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVII	Q	23	230	23	230	23	235	21	219	24	240	ns
AA9-AA16	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVII	Q	45	450	45	450	45	455	43	439	46	460	ns
BB1-BB8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXVIII	Q	23	230	23	230	23	235	21	219	24	240	ns
BB9-BB16	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXVIII	Q	45	450	45	450	45	455	43	439	46	460	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
CC1-CC8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXIX	Q	17	175	17	175	17	179	16	165	18	184	ns
CC9-CC16	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXIX	Q	32	325	32	325	32	329	31	315	34	344	ns
DD1-DD8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXX	Q	17	175	17	175	17	179	16	165	18	184	ns
DD9-DD16	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXX	Q	32	325	32	325	32	329	31	315	34	344	ns
EE1-EE4	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXXI	S	66	660	66	660	67	670	63	636	68	682	ns
EE5-EE8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXXI	S	110	1100	110	1100	111	1110	107	1076	112	1122	ns
EE9-EE12	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXXI	Q	86	860	86	860	89	892	78	788	92	928	ns
EE13-EE16	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXXI	Q	130	1300	130	1300	133	1332	122	1228	136	1368	ns
FF1-FF4	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXXII	S	66	660	66	660	67	670	63	636	68	682	ns
FF5-FF8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXXII	S	110	1100	110	1100	111	1110	107	1076	112	1122	ns
FF9-FF12	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXXII	Q	86	860	86	860	89	892	78	788	92	928	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
FF13-FF16	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXXII	Q	130	1300	130	1300	133	1332	122	1228	136	1368	ns
GG1-GG4	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXXIII	Q	23	230	23	230	23	235	21	219	24	240	ns
GG5-GG8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXXIII	Q	60	600	60	600	60	605	58	589	61	610	ns
HH1-HH4	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXXIV	Q	23	230	23	230	23	235	21	219	24	240	ns
HH5-HH8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXXIV	Q	60	600	60	600	60	605	58	589	61	610	ns
JJ1-JJ4	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXV	Q	17	175	17	175	17	179	16	165	18	184	ns
JJ5-JJ8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXV	Q	50	500	50	500	50	504	49	490	51	509	ns
KK1-KK4	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXVI	Q	17	175	17	175	17	179	16	165	18	184	ns
KK5-KK8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXVI	Q	50	500	50	500	50	504	49	490	51	509	ns
LL1-LL4	Output retriggerability check, V_{DD} = 5 V	XXXVII	X	1.65	6.70	1.59	6.47	1.72	6.97	1.45	5.90	1.78	7.24	µs
MM1-MM4	Output retriggerability check, V_{DD} = 10 V	XXXVIII	S	1.25	7.50	1.21	7.24	1.30	7.80	1.10	6.60	1.35	8.10	µs

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
NN1-NN4	Output retriggerability check, V_{DD} = 15 V	XXXIX	S	1.25	7.50	1.21	7.24	1.30	7.80	1.10	6.60	1.35	8.10	µs
PP1-PP4	Resistance	XL	X	9.9	10.1	9.9	10.1	9.8	10.2	9.9	10.1	9.6	10.4	kΩ
QQ	Power supply filter capacitance	XLI	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
RR1-RR4	Capacitance	XLII	S	198	242	198	242	198	242	198	242	198	242	pF
SS1-SS16	Input capacitance	XLIII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through SS shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through SS in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through J and M through RR shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions K and L shall be performed at 25°, 60° and 100°C only using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition QQ, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through RR shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through SS shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

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	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	18	18	mW
Input current (V_{DD} = 18 V)			
Low level (V_{IL} = 0 V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V)	0 min, 200 max	0 min, -200 max	nA
Output drive current (V_{DD} = 5 V)			
Low level (V_{OL} = 0.4 V)			
I_{OL} at -55°C	---	0.64 min	mA
0° and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
100°C	---	0.36 min	mA
High level (V_{OH} = 4.6 V)			
I_{OH} at -55°C	---	-0.25 min	mA
0° and 25°C	-0.20 min	-0.20 min	mA
60°C	-0.17 min	-0.17 min	mA
100°C	---	-0.14 min	mA

6.5 Module function. The monostable multivibrator module contains 4 independent monostable multivibrator circuits. The module provides stable retriggerable, resettable one shot operation for any fixed voltage timing operation. An external resistor (R_X) and an external capacitor (C_X) control the timing for the circuit. Adjustment of R_X and C_X provides a wide range of output pulse widths from the Q and $\overline{Q}$ outputs. The minimum value of R_X which can be used is 5 kΩ while the maximum is 1 MΩ. Leading edge triggering (A) and trailing edge triggering (B) inputs are provided for triggering from either edge of an input pulse. A reset (on low level) is provided for immediate termination of the output pulse or to prevent output pulses when power is turned on. There are also four 220 pF capacitors, with one side connected to ground and four 10 kΩ resistors, with one side connected to V_{DD}, for external use.

6.5.1 Input to output relationship. Each monostable multivibrator circuit shall have the input to output relationship of table II.

TABLE II. Input to output relationship

Input					Output	
A	B	R	T1	T2	Q	$\overline{Q}$
H	L	L	X	X	L	H
H	L	H	X	L	H	L

L = low level

H = high level

X = don't care

6.5.2 Parallel outputs. The paralleling of identical outputs on integrated circuits is permissible.

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6.5.3 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{TOTAL} = (C\ V_{DD}2f)\ N + \text{Quiescent power dissipation.}$$

Where: C = load capacitance.
V_{DD} = supply voltage.
f = operating frequency.
N = number of outputs.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply is off.

d. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.

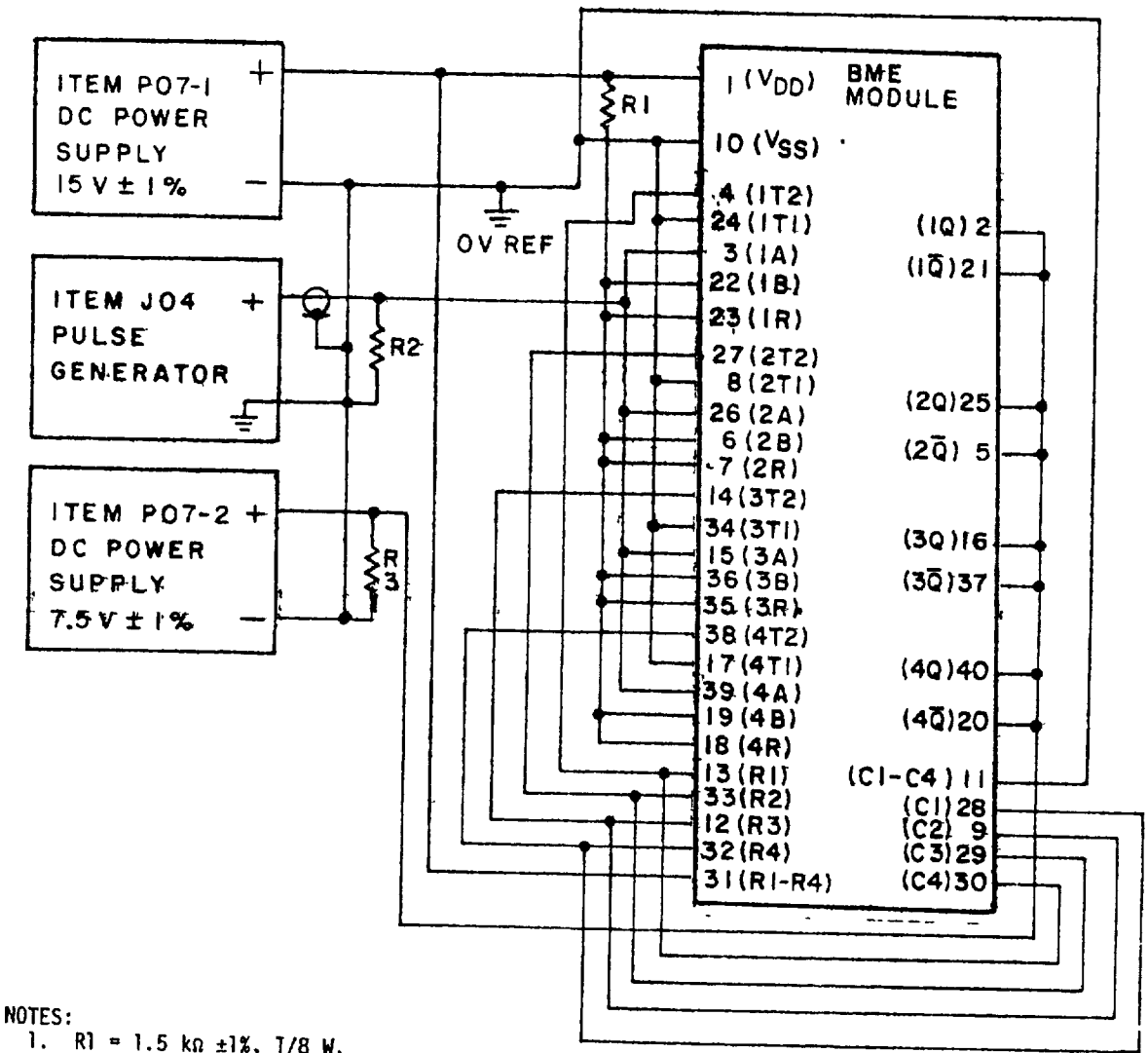
e. All unused input leads must be connected to either V_{SS} (GND) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

Agent
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N302)

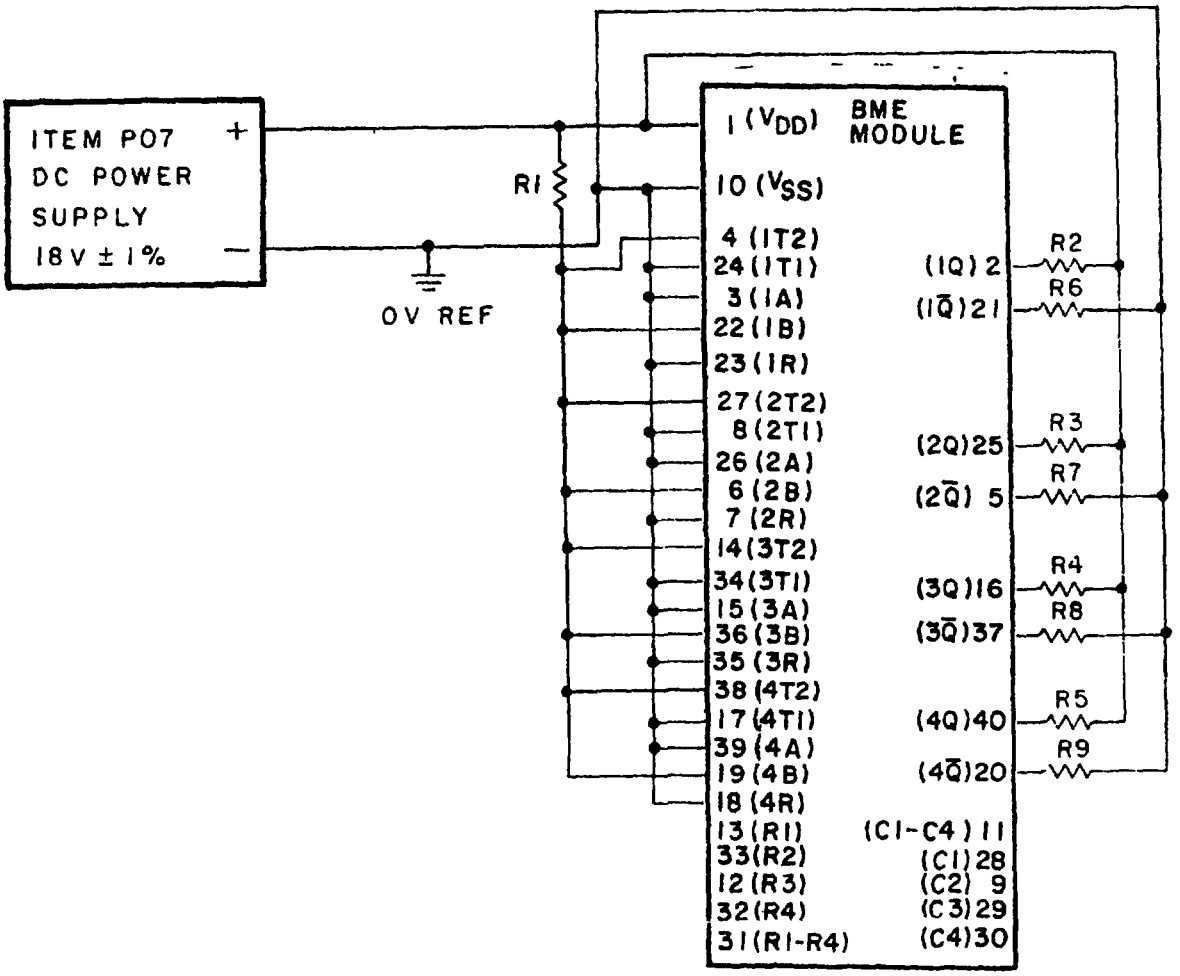
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/8 W.
2. R2 = R3 = 51.1 Ω ±5%, 5 W.
3 Item J04 setup.
PW = 500 ±100 ns.
PA = $15\ ^{+1}_{-0}$ V.
$t_{TLH} = t_{THL}$ = 50 ±10 ns
Frequency = 1 MHz ±20%

FIGURE 2. Component temperature test circuit.

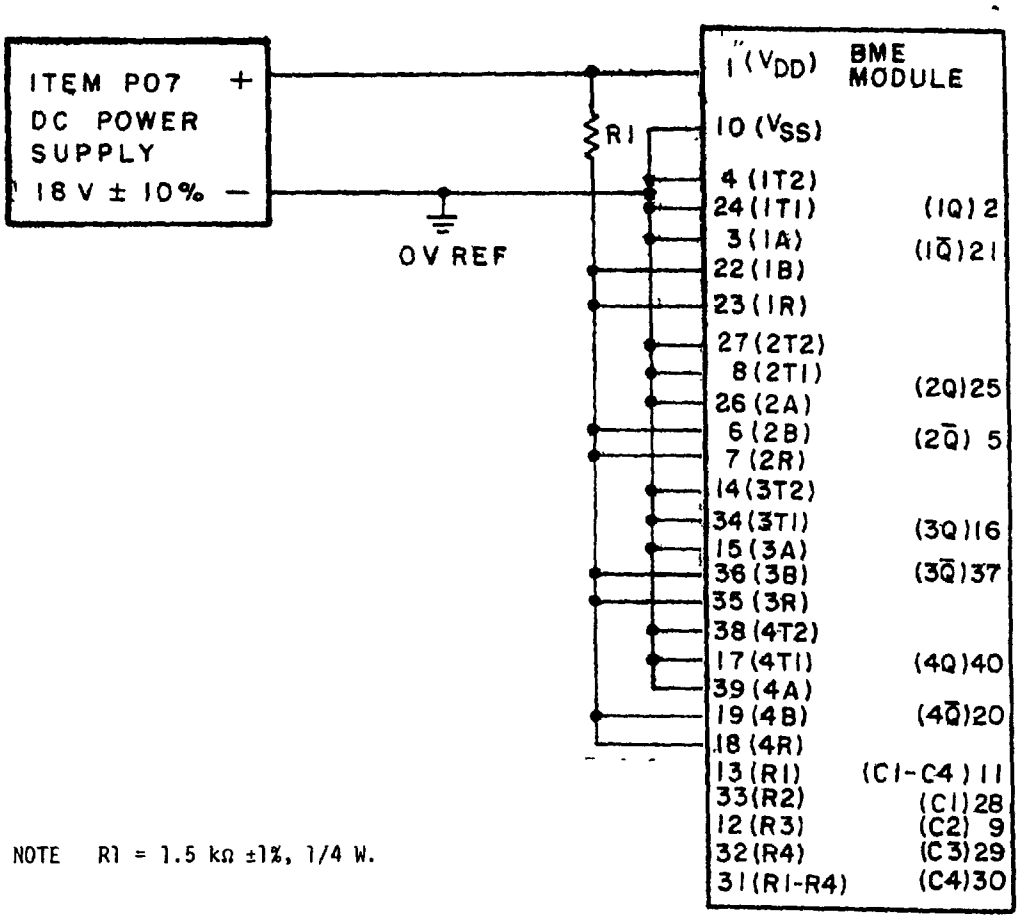
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NOTES.
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2-R5 = 5.62 kΩ ±1%, 1/8 W.
3. R6-R9 = 12.1 kΩ ±1%, 1/8 W.

FIGURE 3. Operating temperature and life test circuit.

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NOTE R1 = 1.5 kΩ ±1%, 1/4 W.

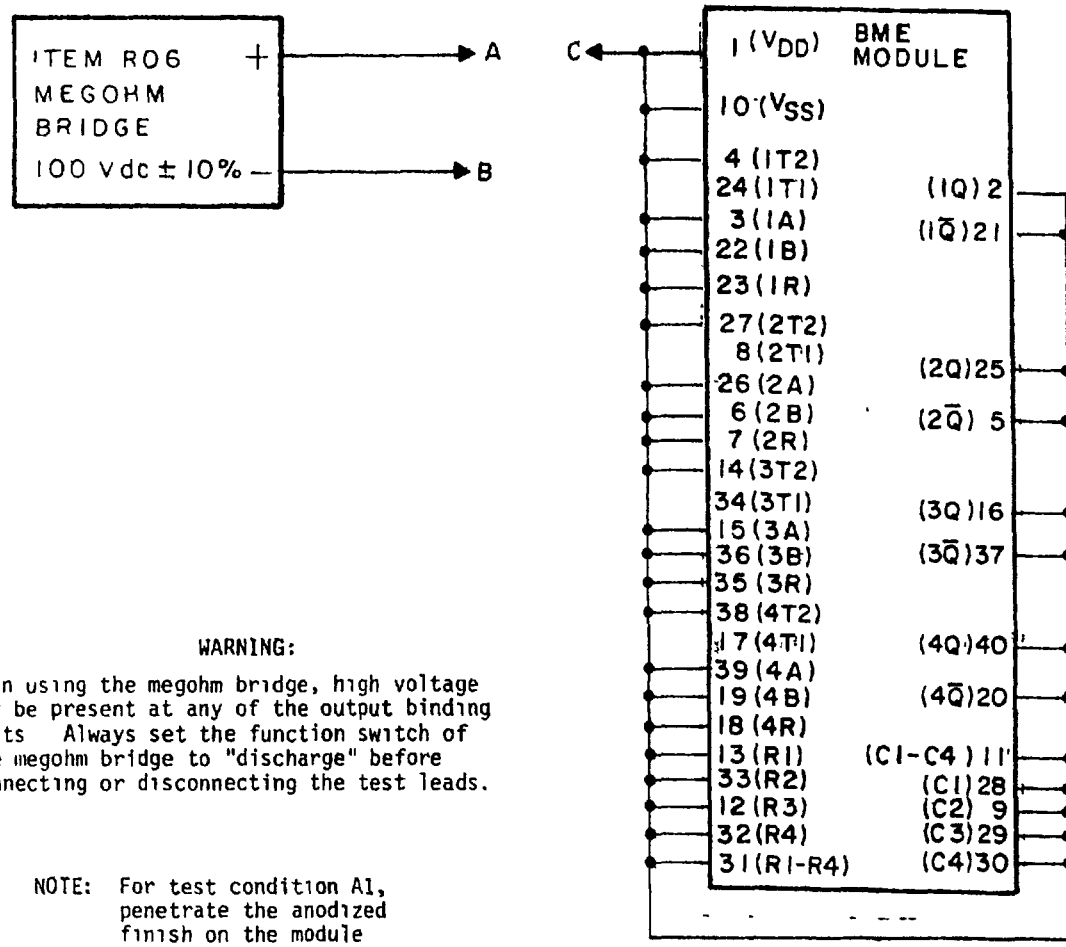
FIGURE 4 Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C,24,8,34,17	Frame	10	∞	MΩ	5
A2	Isolation	C	24,8,34,17	10	∞	MΩ	5

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WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Input clamp diode voltage (negative) test.

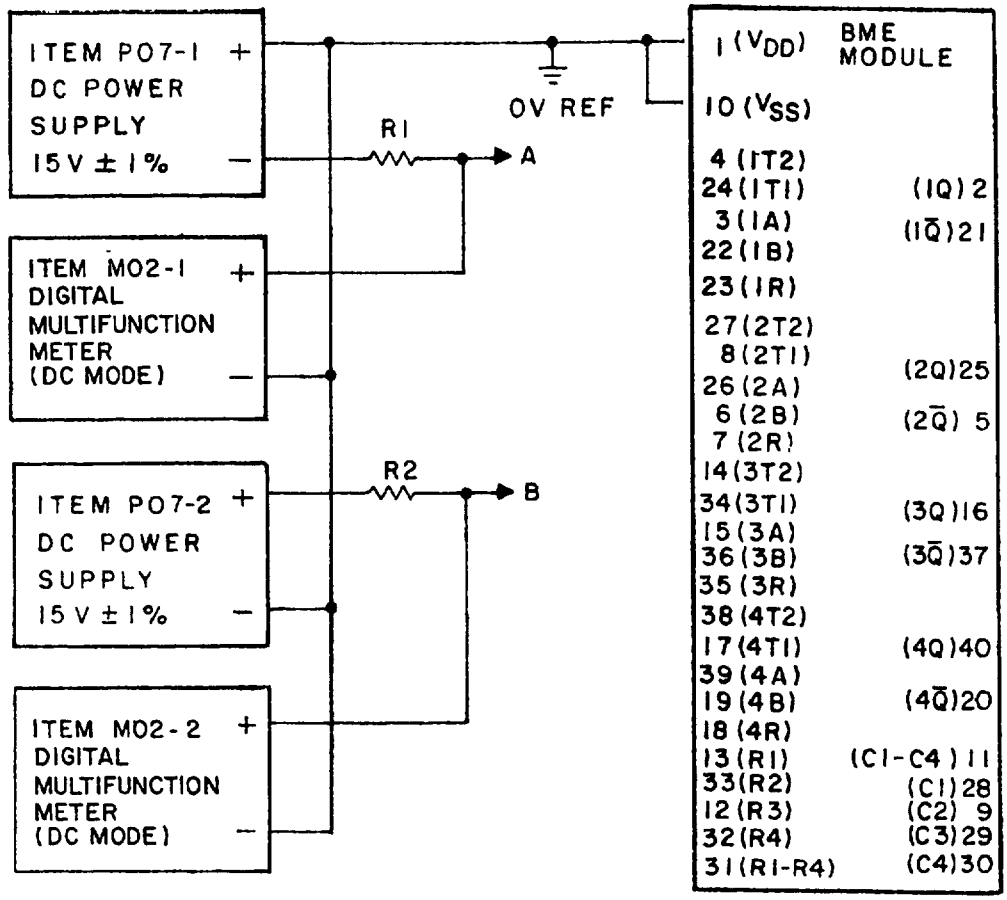
Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	1T2 Input clamp diode voltage (negative)	4	Open	-0.4	-6.0	V	6
B2	1A	3					
B3	1B	22					
B4	1R	23					
B5	2T2	27					
B6	2A	26					
B7	2B	6					
B8	2R	7					
B9	3T2	14					
B10	3A	15					
B11	3B	36					
B12	3R	35					
B13	4T2	38					
B14	4A	39					
B15	4B	19					
B16	4R	18					

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TABLE V. Input clamp diode voltage (positive) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1T2 Input clamp diode voltage (positive)	Open	4	0.4	1.5	V	6
C2	1A		3				
C3	1B		22				
C4	1R		23				
C5	2T2		27				
C6	2A		26				
C7	2B		6				
C8	2R		7				
C9	3T2		14				
C10	3A		15				
C11	3B		36				
C12	3R		35				
C13	4T2		38				
C14	4A		39				
C15	4B		19				
C16	4R		18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:08Z
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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items PO7-1, PO7-2, MO2-1 and MO2-2 are not used simultaneously, only one item PO7 and one item MO2 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

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TABLE VI. Output voltage (low-level, I_{OL} = 0) test.

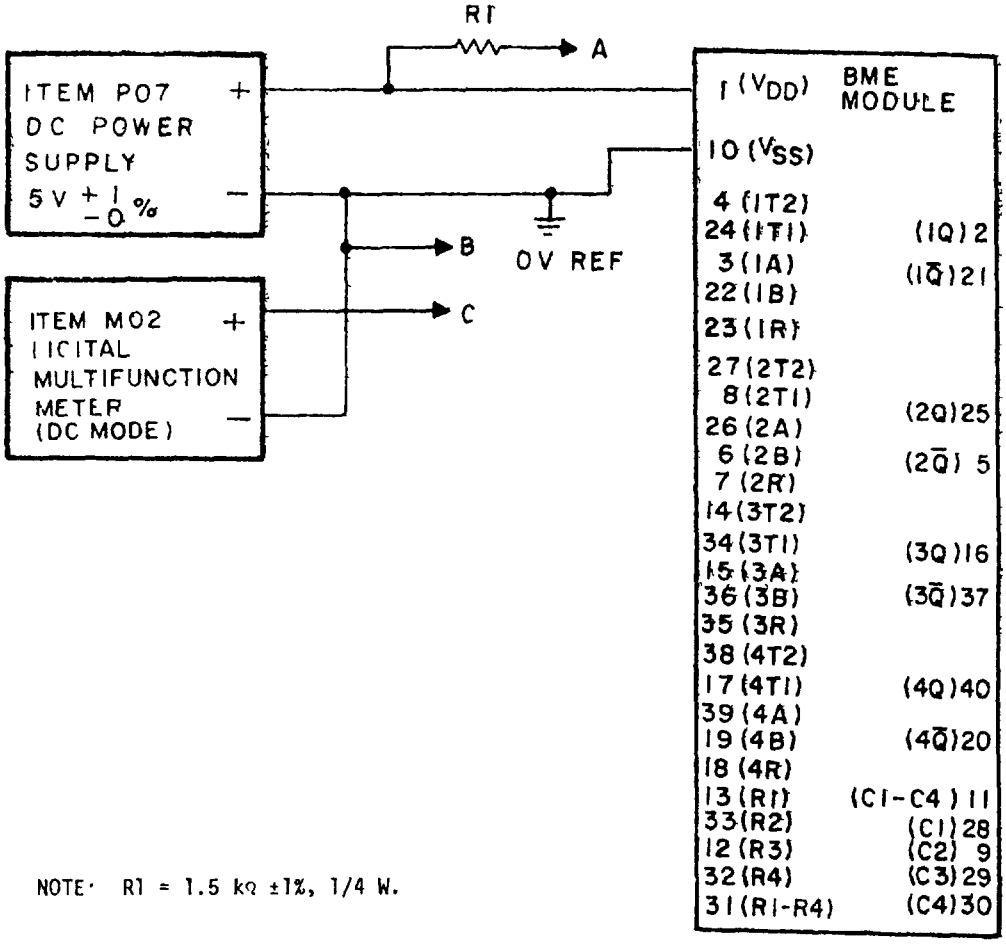
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	1Q	Output voltage, low level (I_{OL} = 0)	3,26,15,39	22,23,6,7,36, 35,19,18	2	0	50	mV	7
D2	2Q				25				
D3	3Q				16				
D4	4Q				40				
D5	$1\overline{Q}$		23,7,35,18, 3,26,15,39	4,27,14,38,22, 6,36,19	21				
D6	$2\overline{Q}$				5				
D7	$3\overline{Q}$				37				
D8	$4\overline{Q}$				20				

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TABLE VII. Output voltage (high-level, I_{OH} = 0) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	1Q	Output voltage, high level (I_{OH} = 0)	23,7,35,18, 3,26,15,39	4,27,14,38,22, 6,36,19	2	4.95	5.05	V	7
E2	2Q				25				
E3	3Q				16				
E4	4Q				40				
E5	$1\overline{Q}$		3,26,15,19	22,23,6,7,36, 35,19,18	21				
E6	$2\overline{Q}$				5				
E7	$3\overline{Q}$				37				
E8	$4\overline{Q}$				20				

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NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

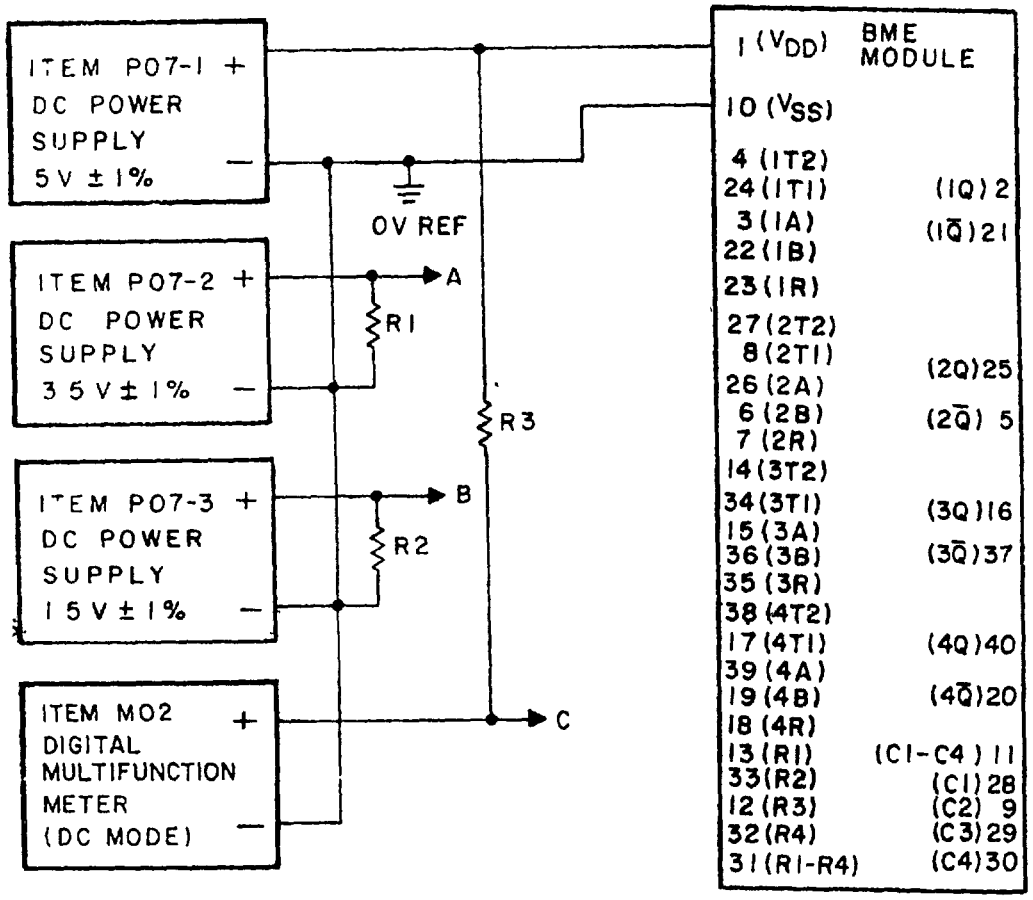
FIGURE 7. Output voltage (low-level, I_{OL} = 0 and high-level, I_{OH} = 0) test circuit for tables VI and VII.

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TABLE VIII. Output voltage (low-level, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1Q Output voltage, low level (V_{DD} = 5 V)	3,7,35,18	22,23,27,14,38,26,6,15,36,39,19	2	0.0	0.4	V	8
F2	2Q	26,23,35,18	6,7,4,14,38,3,22,15,36,39,19	25				
F3	3Q	15,23,7,18	36,35,4,27,38,3,22,26,6,39,19	16				
F4	4Q	39,23,7,35	19,18,4,27,14,3,22,26,6,15,36	40				
F5	$1\overline{Q}$	23,3	4,22,7,35,18,26,6,15,36,39,19	21				
F6	$2\overline{Q}$	7,26	27,6,23,35,18,3,22,15,36,39,19	5				
F7	$3\overline{Q}$	35,15	14,36,23,7,18,3,22,26,6,39,19	37				
F8	$4\overline{Q}$	18,39	38,19,23,7,38,3,22,26,6,15,36	20				

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NOTES
1. R1 = R2 = 51.1 Ω ±5%, 1/2 W.
2. R3 = 7.15 kΩ ±1%, 1/8 W at -55°C.
= 9.09 kΩ ±1%, 1/8 W at 0° and 25°C.
= 10.5 kΩ ±1%, 1/8 W at 60°C.
= 12.7 kΩ ±1%, 1/8 W at 100°C.

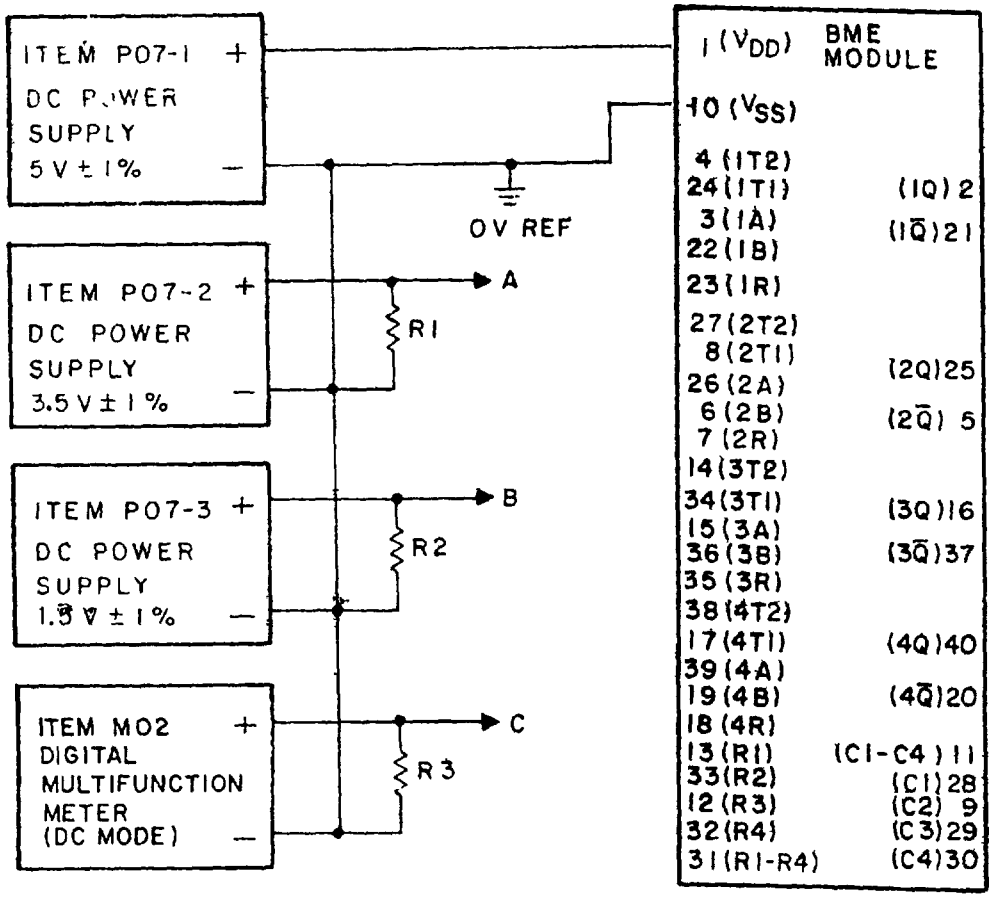
FIGURE 8. Output voltage (low-level, V_{DD} = 5 V) test circuit for table VIII.

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TABLE IX. Output voltage (high-level, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1Q Output voltage, high level (V_{DD} = 5 V)	23,3	4,22,7,35,18,26,6,15,36,39,19	2	4.6	5.0	V	9
G2	2Q	7,26	27,6,23,35,18,3,22,15,36,39,19	25				
G3	3Q	35,15	14,36,23,7,18,3,22,26,6,39,19	16				
G4	4Q	18,39	38,19,23,7,35,3,22,26,6,15,36	40				
G5	$1\overline{Q}$	3,7,35,18	22,23,27,14,38,26,6,15,36,39,19	21				
G6	$2\overline{Q}$	26,23,35,18	6,7,4,14,38,3,22,15,36,39,19	5				
G7	$3\overline{Q}$	15,23,7,18	36,35,4,27,38,3,22,26,6,39,19	37				
G8	$4\overline{Q}$	39,23,7,35	19,18,4,27,14,3,22,26,6,15,36	20				

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NOTES:
1. R1 = R2 = 51.1 Ω ±5%, 1/2 W.
2. R3 = 18.2 kΩ ±1%, 1/8 W at -55°C.
= 23.2 kΩ ±1%, 1/8 W at 0° and 25°C.
= 27.4 kΩ ±1%, 1/8 W at 60°C.
= 32.4 kΩ ±1%, 1/8 W at 100°C.

FIGURE 9. Output voltage (high-level, V_{DD} = 5 V) test circuit for table IX.

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TABLE X. Output voltage (low-level, V_{DD} = 15 V) test.

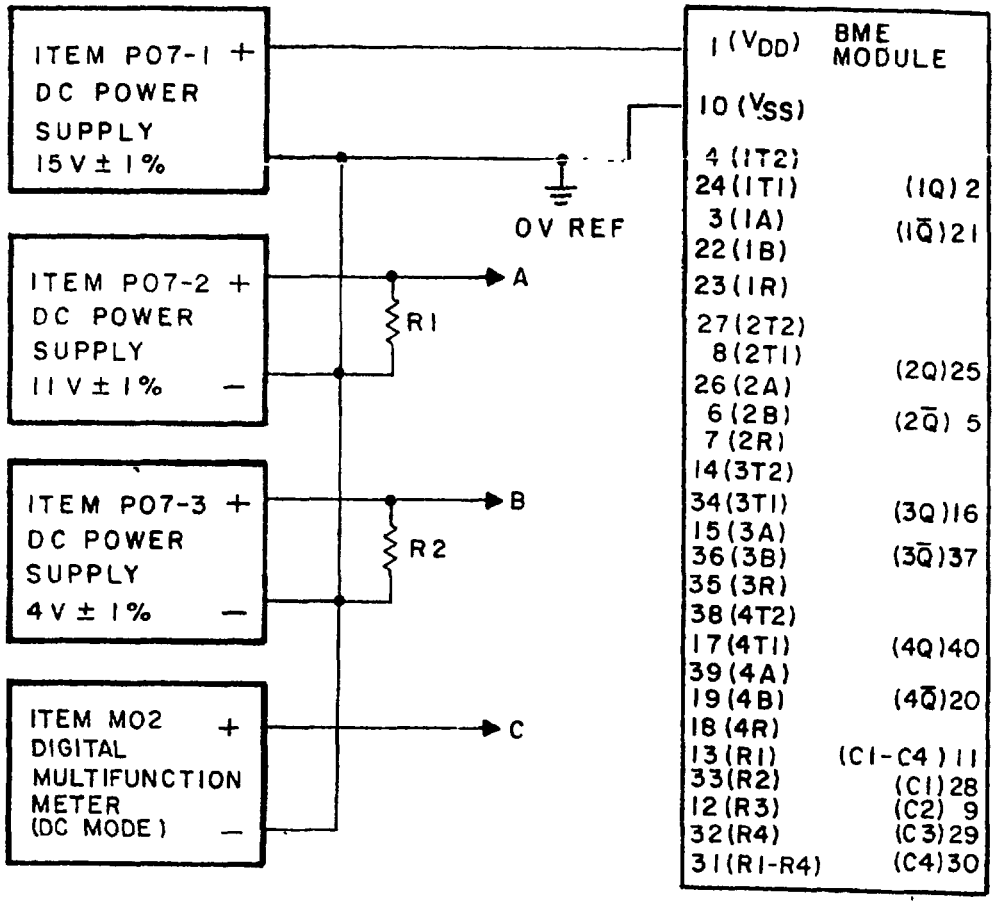
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1Q Output voltage, low level (V_{DD} = 15 V)	3,26,15,39	22,23,6,7,36,35,19,18	2	0.0	1.5	V	10
H2	2Q			25				
H3	3Q			16				
H4	4Q			40				
H5	$1\overline{Q}$	23,7,35,18, 3,26,15,39	4,27,14,38,22,6,36,19	21				
H6	$2\overline{Q}$			5				
H7	$3\overline{Q}$			37				
H8	$4\overline{Q}$			20				

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TABLE XI. Output voltage (high-level, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1Q Output voltage, high level (V_{DD} = 15 V)	23,7,35,18, 3,26,15,39	4,27,14,38,22,6,36,19	2	13.5	15.0	V	10
J2	2Q			25				
J3	3Q			16				
J4	4Q			40				
J5	$1\overline{Q}$	3,26,15,39	22,23,6,7,36,35,19,18	21				
J6	$2\overline{Q}$			5				
J7	$3\overline{Q}$			37				
J8	$4\overline{Q}$			20				

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NOTES:
1. R1 = 576 Ω ±5%, 1/4 W.
2. R2 = 51.1 Ω ±5%, 1/2 W.

FIGURE 10. Output voltage (low-level and high-level, V_{DD} = 15 V) test circuit for tables X and XI.

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TABLE XII. Input current (low-level) test.

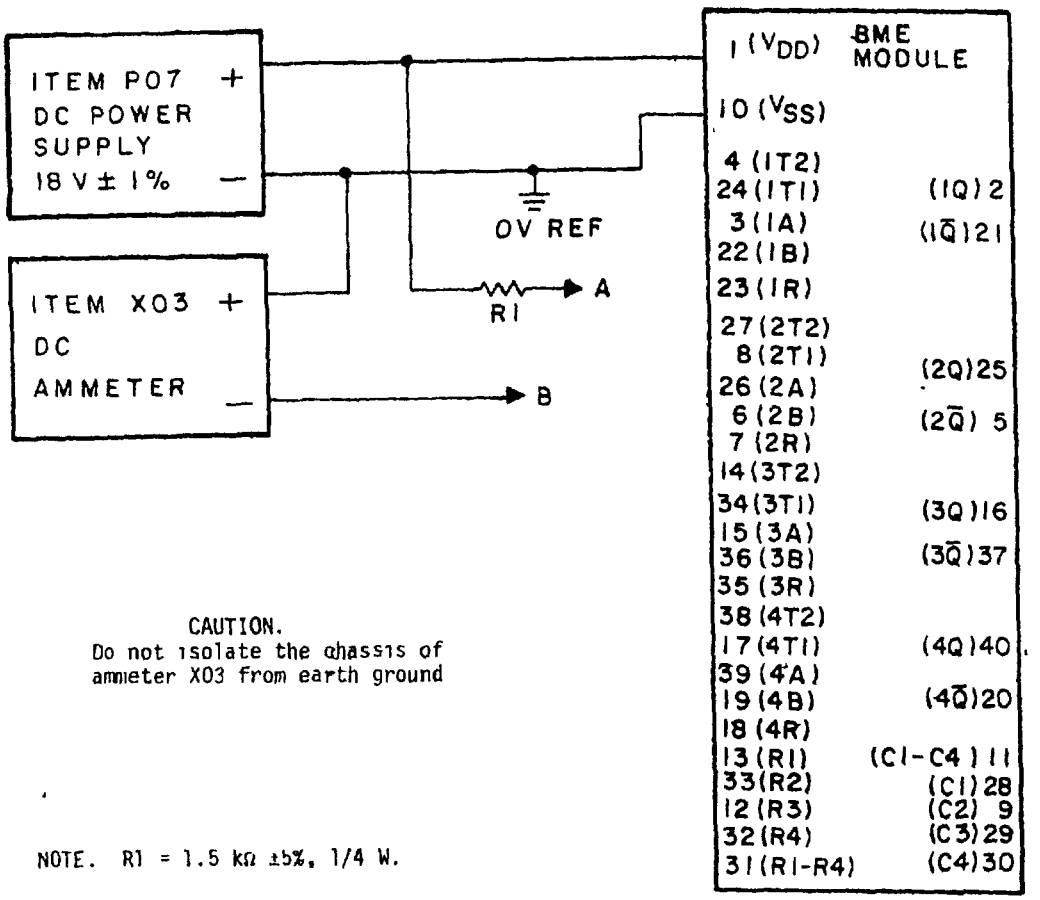
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1T2 Input current, low level	24,3,22,23,8,27,26,6,7,34,14,15,36,35, 17,38,39,19,18	4	0	-200	nA	11
K2	1A	24,4,22,23,8,27,26,6,7,34,14,15,36,35, 17,38,39,19,18	3				
K3	1B	24,4,3,23,8,27,26,6,7,34,14,15,36,35, 17,38,39,19,18	22				
K4	1R	24,4,3,22,8,27,26,6,7,34,14,15,36,35, 17,38,39,19,18	23				
K5	2T2	24,4,3,22,23,8,26,6,7,34,14,15,36,35, 17,38,39,19,18	27				
K6	2A	24,4,3,22,23,8,27,6,7,34,14,15,36,35, 17,38,39,19,18	26				
K7	2B	24,4,3,22,23,8,27,26,7,34,14,15,36,35, 17,38,39,19,18	6				
K8	2R	24,4,3,22,23,8,27,26,6,34,14,15,36,35, 17,38,39,19,18	7				
K9	3T2	24,4,3,22,23,8,27,26,6,7,34,15,36,35, 17,38,39,19,18	14				
K10	3A	24,4,3,22,23,8,27,26,6,7,34,14,36,35, 17,38,39,19,18	15				
K11	3B	24,4,3,22,23,8,27,26,6,7,34,14,15,35, 17,38,39,19,18	36				
K12	3R	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 17,38,39,19,18	35				

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K13	4T2 Input current, low level	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 35,17,39,19,18	38	0	-200	nA	11
K14	4A	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 35,17,38,19,18	39				
K15	4B	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 35,17,38,39,18	19				
K16	4R	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 35,17,38,39,19	18				

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CAUTION.
Do not isolate the chassis of ammeter X03 from earth ground

NOTE. R1 = 1.5 kΩ ±5%, 1/4 W.

FIGURE 11. Input current (low-level) test circuit for table XII.

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TABLE XIII. Input current (high-level) test.

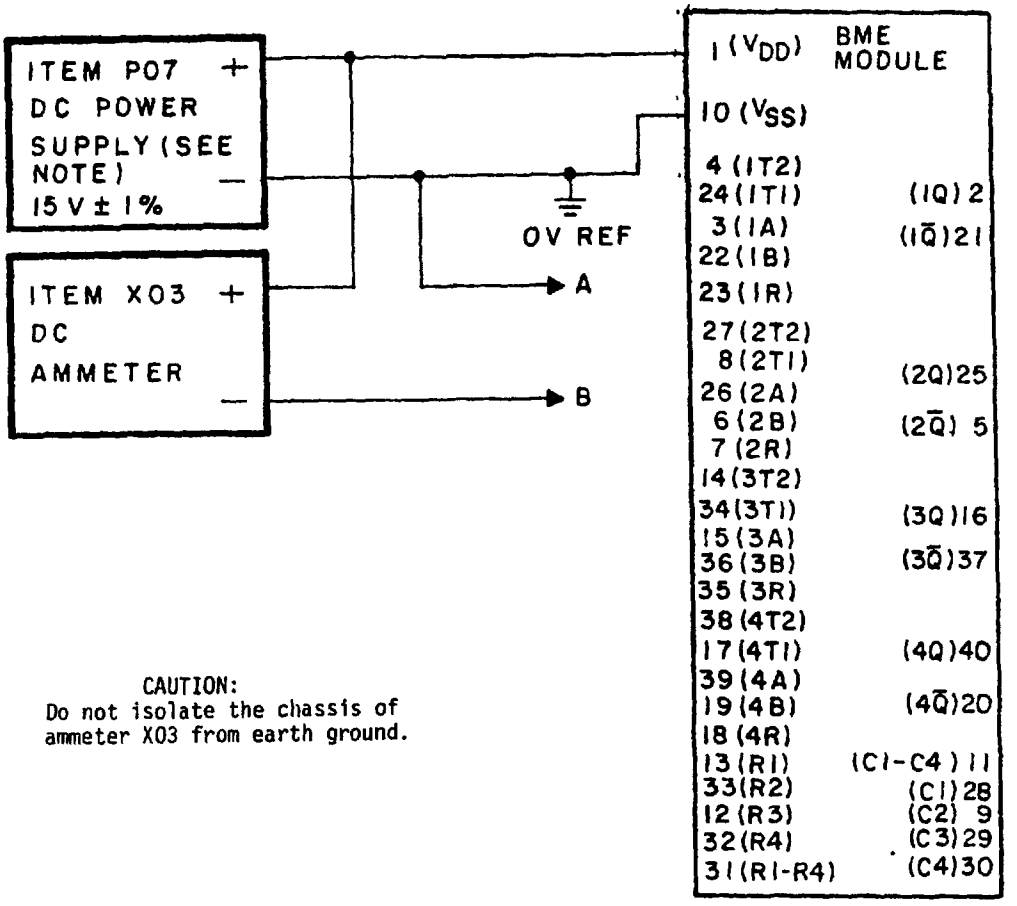
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1T2 Input current, high level	24,3,22,23,8,27,26,6,7,34,14,15,36,35, 17,38,39,19,18	4	0	200	nA	12
L2	1A	24,4,22,23,8,27,26,6,7,34,14,15,36,35, 17,38,39,19,18	3				
L3	1B	24,4,3,23,8,27,26,6,7,34,14,15,36,35, 17,38,39,19,18	22				
L4	1R	24,4,3,22,8,27,26,6,7,34,14,15,36,35, 17,38,39,19,18	23				
L5	2T2	24,4,3,22,23,8,26,6,7,34,14,15,36,35, 17,38,39,19,18	27				
L6	2A	24,4,3,22,23,8,27,6,7,34,14,15,36,35, 17,38,39,19,18	26				
L7	2B	24,4,3,22,23,8,27,26,7,34,14,15,36,35, 17,38,39,19,18	6				
L8	2R	24,4,3,22,23,8,27,26,6,34,14,15,36,35, 17,38,39,19,18	7				
L9	3T2	24,4,3,22,23,8,27,26,6,7,34,15,36,35, 17,38,39,19,18	14				
L10	3A	24,4,3,22,23,8,27,26,6,7,34,14,36,35, 17,38,39,19,18	15				
L11	3B	24,4,3,22,23,8,27,26,6,7,34,14,15,35, 17,38,39,19,18	36				
L12	3R	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 17,38,39,19,18	35				

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L13	4T2 Input current, high level	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 35,17,39,19,18	38	0	200	nA	12
L14	4A	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 35,17,38,19,18	39	↓	↓	↓	↓
L15	4B	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 35,17,38,39,18	19				
L16	4R	24,4,3,22,23,8,27,26,6,7,34,14,15,36, 35,17,38,39,19	18				

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CAUTION:
Do not isolate the chassis of ammeter X03 from earth ground.

NOTE: Limit item P07 current to 1.0 mA.

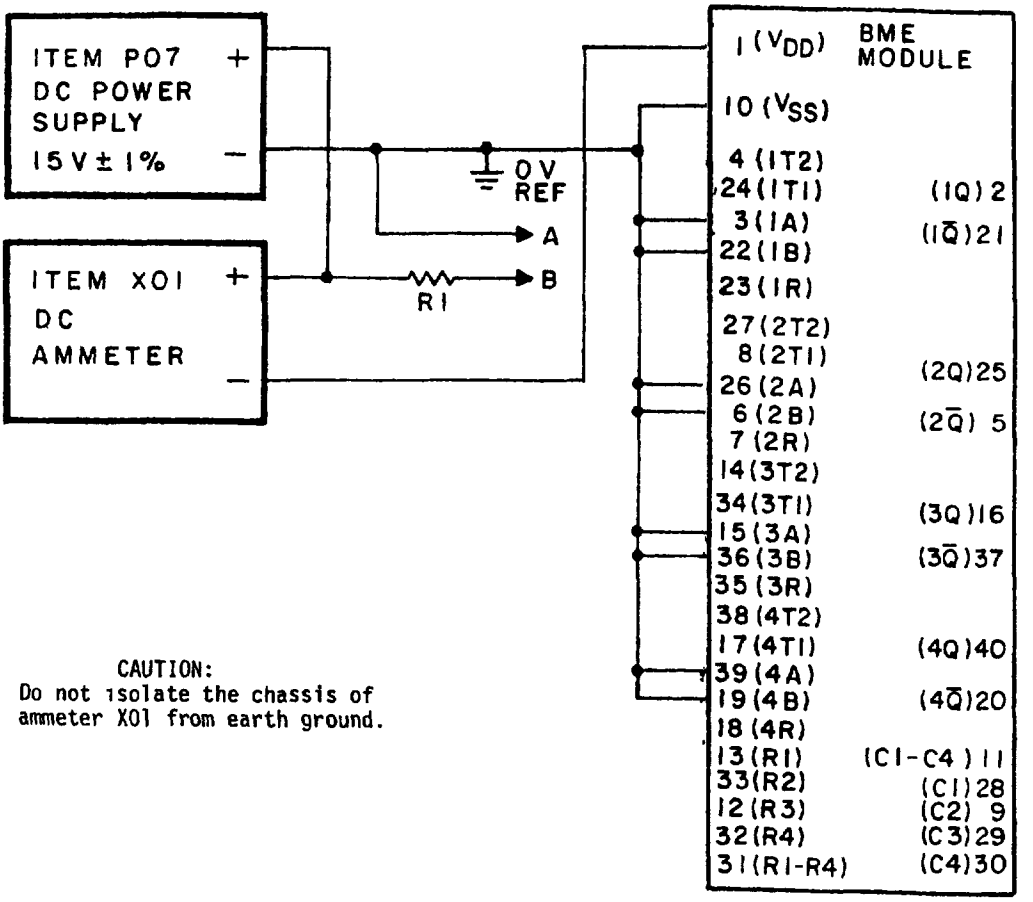
FIGURE 12. Input current (high-level) test circuit for table XIII

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TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level	23,7,35,18	Open	0	40	µA	13
M2	Power supply current, high level	4,27,14,38	23,7,35,18	0	40	µA	13

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 13. Power supply current (low-level and high-level) test circuit for table XIV.

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TABLE XV. Output pulse width (match, V_{DD} = 5 V) test.

Test condition	Parameter	Connect									Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Pin 13 to pin	Pin 33 to pin	Pin 28 to pin	Pin 9 to pin	Min	Max		
N1	1Q to 2Q, Output pulse width, match, V_{DD} = 5 V	3,26	2	25	Open	Open	4	27	4	27	-400	400	ns	14,15
N2	3Q to 4Q	15,39	16	40	Open	Open	14	38	14	38	-400	400	ns	
N3	1Q to 2Q	3,26	2	25	4	27	Open	Open	Open	Open	-55	55	µs	
N4	3Q to 4Q	15,39	16	40	14	38	Open	Open	Open	Open	-55	55	µs	

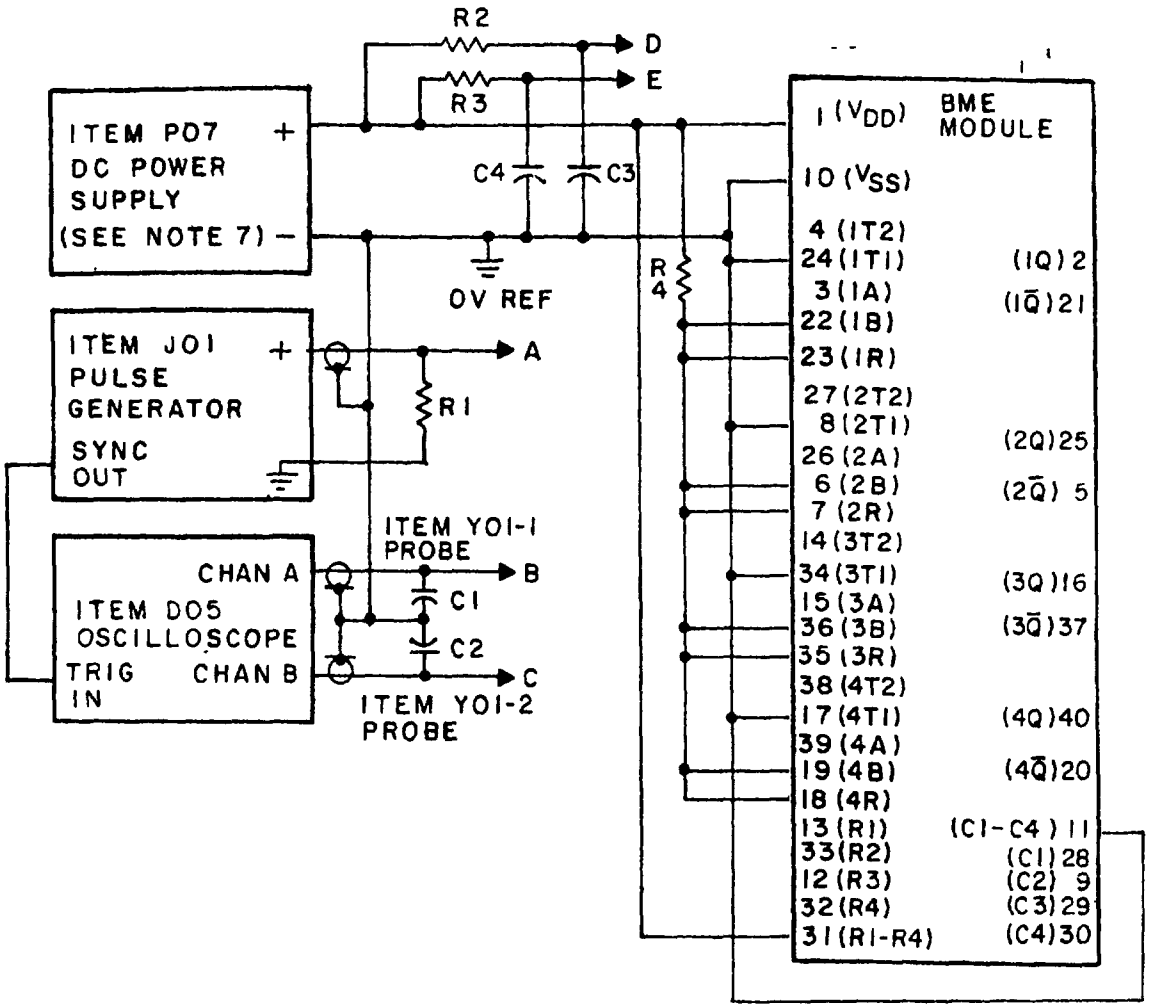
TABLE XVI. Output pulse width (match, V_{DD} = 10 V) test.

Test condition	Parameter	Connect									Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Pin 13 to pin	Pin 33 to pin	Pin 28 to pin	Pin 9 to pin	Min	Max		
P1	1Q to 2Q, Output pulse width, match, V_{DD} = 10 V	3,26	2	25	Open	Open	4	27	4	27	-500	500	ns	14,15
P2	3Q to 4Q	15,39	16	40	Open	Open	14	38	14	38	-500	500	ns	
P3	1Q to 2Q	3,26	2	25	4	27	Open	Open	Open	Open	-110	110	µs	
P4	3Q to 4Q	15,39	16	40	14	38	Open	Open	Open	Open	-110	110	µs	

TABLE XVII. Output pulse width (match, V_{DD} = 15 V) test.

Test condition	Parameter	Connect									Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Pin 13 to pin	Pin 33 to pin	Pin 28 to pin	Pin 9 to pin	Min	Max		
Q1	1Q to 2Q, Output pulse width, match, V_{DD} = 15 V	3,26	2	25	Open	Open	4	27	4	27	-500	500	ns	14,15
Q2	3Q to 4Q	15,39	16	40	Open	Open	14	38	14	38	-500	500	ns	
Q3	1Q to 2Q	3,26	2	25	4	27	Open	Open	Open	Open	-130	130	µs	
Q4	3Q to 4Q	15,39	16	40	14	38	Open	Open	Open	Open	-130	130	µs	

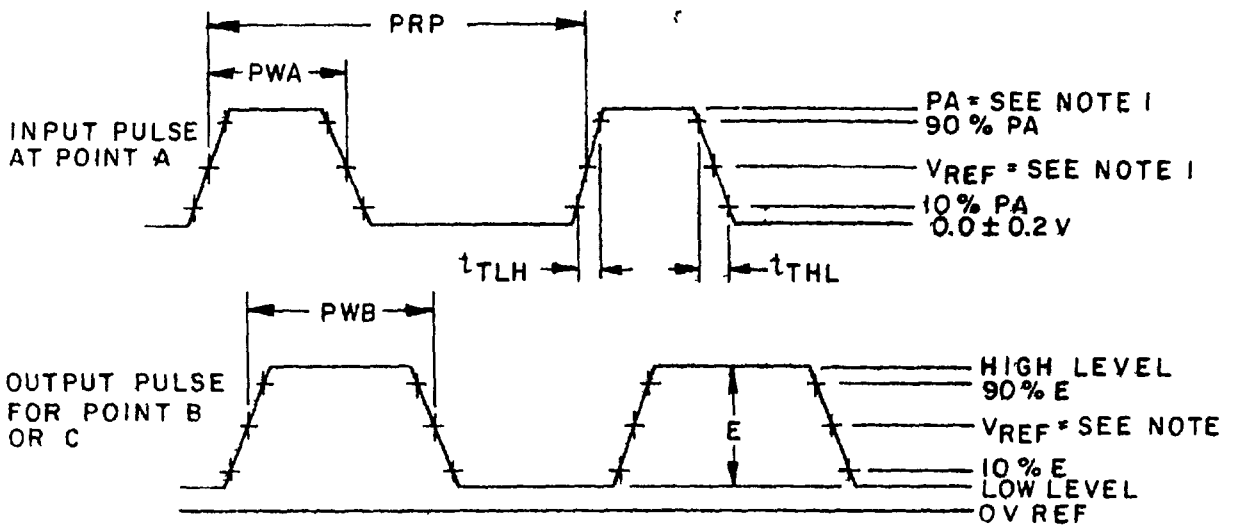
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NOTES:
1. R1 = 51.1 Ω ±5%, 1/2 W.
2. R2 = R3 100 kΩ ±1%, 1/8 W.
3. R4 = 1.5 kΩ ±1%, 1/4 W.
4. C1 = C2 = 50 ±5 pF (including probe and parasitic capacitance).
5. C3 = C4 = 0.01 µF ±5%.
6. Take the readings at points B and C. Then take the readings from point C and subtract it from point B and the differential should be in the limits as shown in tables.
7. Set item PO7 as follows:
   5.0 V ±1% for test conditions N1-N4.
   10.0 V ±1% for test conditions P1-P4.
   15.0 V ±1% for test conditions Q1-Q4.

FIGURE 14. Output pulse width (match) test circuit for tables XV, XVI and XVII.

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NOTES:

1. Input pulse parameters:
   PA = 5.0 ±0.1 V for test conditions N1-N4.
   = 10.0 ±0.2 V for test conditions P1-P4.
   = 15.0 ±0.3 V for test conditions Q1-Q4.
   V_{REF} = 2.5 ±0.1 V for test conditions N1-N4.
   = 5.0 ±0.2 V for test conditions P1-P4.
   = 7.5 ±0.3 V for test conditions Q1-Q4.
   PWA = 1.0 ±0.1 µs for test conditions N1, N2, P1, P2, Q1, Q2.
   = 150 µs ±10% for test conditions N3, N4, P3, P4, Q3, Q4
   PRP = 5.0 ±0.5 µs for test conditions N1, N2, P1, P2, Q1, Q2.
   = 600 µs ±10% for test conditions N3, N4.
   = 900 µs ±10% for test conditions P3, P4.
   =1100 µs ±10% for test conditions Q3, Q4
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows
   High level ≥ 4.6 V for test conditions: N1-N4.
   ≥ 9.5 V for test conditions: P1-P4.
   ≥ 13.5 V for test conditions: Q1-Q4.
   Low level ≤ 0.4 V for test conditions. N1-N4.
   ≤ 0.5 V for test conditions: P1-P4.
   ≤ 1.5 V for test conditions: Q1-Q4.
3. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probes YO1-1 and YO1-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground (V_{SS})

FIGURE 15. Output pulse width (match) test circuit waveforms for tables XV, XVI, and XVII.

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TABLE XVIII. Output pulse width check (V_{DD} = 5 V) test.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Pin 13 to pin	Pin 28 to pin	Pin 10 to pin	Min	Max		
R1	1Q Output pulse width check (V_{DD} = 5 V)	3	2	Open	Open	Open	4	4	27,26,14,15,38,39	1.65	3.35	µs	16,17
R2	2Q	26	25				27	27	4,3,14,15,38,39				
R3	3Q	15	16				14	14	4,3,27,26,38,39				
R4	4Q	39	40				38	38	4,3,27,26,14,15				
R5	1Q	3	2		4		4	Open	27,26,14,15,38,39	4.65	9.35		
R6	2Q	26	25		27		27		4,3,14,15,38,39				
R7	3Q	15	16		14		14		4,3,27,26,38,39				
R8	4Q	39	40		38		38		4,3,27,26,14,15				
R9	1Q	3	2		Open	4	4		27,26,14,15,38,39	23.3	46.7		
R10	2Q	26	25			27	27		4,3,14,15,38,39				
R11	3Q	15	16			14	14		4,3,27,26,38,39				
R12	4Q	39	40			38	38		4,3,27,26,14,15				
R13	1Q	3	2	4		4	Open		27,26,14,15,38,39	235	470		
R14	2Q	26	25	27		27			4,3,14,15,38,39				
R15	3Q	15	16	14		14			4,3,27,26,38,39				
R16	4Q	39	40	38		38			4,3,27,26,14,15				

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TABLE XIX. Output pulse width check (V_{DD} = 10 V) test.

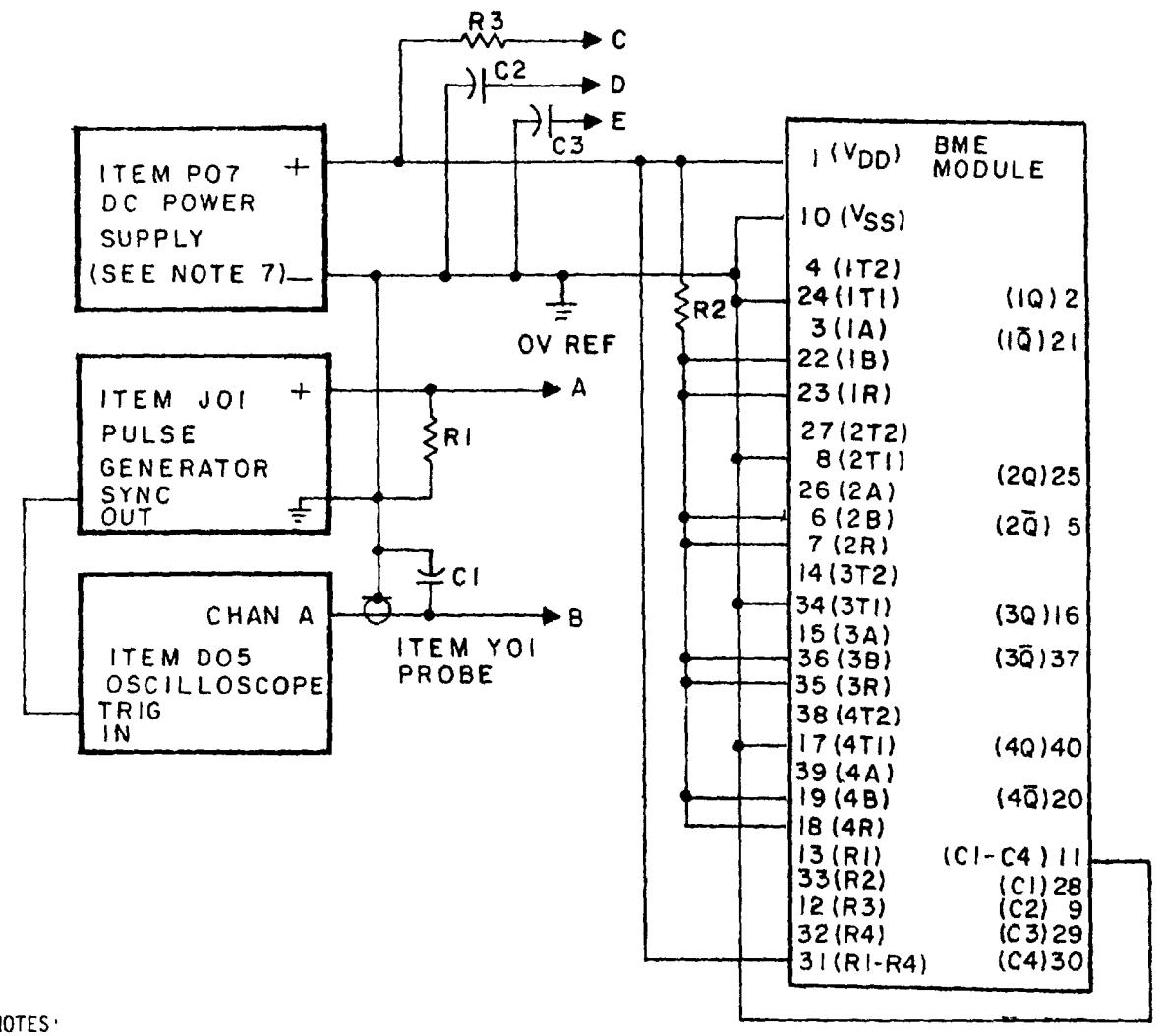
Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Pin 13 to pin	Pin 28 to pin	Pin 10 to pin	Min	Max		
S1	1Q Output pulse width check (V_{DD} = 10 V)	3	2	Open	Open	Open	4	4	27,26,14,15,38,39	1.25	3.75	µs	16,17
S2	2Q	26	25				27	27	4,3,14,15,38,39				
S3	3Q	15	16				14	14	4,3,27,26,38,39				
S4	4Q	39	40				38	38	4,3,27,26,14,15				
S5	1Q	3	2		4		4	Open	27,26,14,15,38,39	3.75	11.3		
S6	2Q	26	25		27		27		4,3,14,15,38,39				
S7	3Q	15	16		14		14		4,3,27,26,38,39				
S8	4Q	39	40		38		38		4,3,27,26,14,15				
S9	1Q	3	2		Open	4	4		27,26,14,15,38,39	25	75		
S10	2Q	26	25			27	27		4,3,14,15,38,39				
S11	3Q	15	16			14	14		4,3,27,26,38,39				
S12	4Q	39	40			38	38		4,3,27,26,14,15				
S13	1Q	3	2	4		4	Open		27,26,14,15,38,39	275	825		
S14	2Q	26	25	27		27			4,3,14,15,38,39				
S15	3Q	15	16	14		14			4,3,27,26,38,39				
S16	4Q	39	40	38		38			4,3,27,26,14,15				

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TABLE XX. Output pulse width check (V_{DD} = 15 V) test.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Pin 13 to pin	Pin 28 to pin	Pin 10 to pin	Min	Max		
T1	1Q Output pulse width check (V_{DD} = 15 V)	3	2	Open	Open	Open	4	4	27,26,14,15,38,39	1.25	3.75	µs	16,17
T2	2Q	26	25				27	27	4,3,14,15,38,39				
T3	3Q	15	16				14	14	4,3,27,26,38,39				
T4	4Q	39	40				38	38	4,3,27,26,14,15				
T5	1Q	3	2		4		4	Open	27,26,14,15,38,39	4	12		
T6	2Q	26	25		27		27		4,3,14,15,38,39				
T7	3Q	15	16		14		14		4,3,27,26,38,39				
T8	4Q	39	40		38		38		4,3,27,26,14,15				
T9	1Q	3	2		Open	4	4		27,26,14,15,38,39	30	90		
T10	2Q	26	25			27	27		4,3,14,15,38,39				
T11	3Q	15	16			14	14		4,3,27,26,38,39				
T12	4Q	39	40			38	38		4,3,27,26,14,15				
T13	1Q	3	2	4		4	Open		27,26,14,15,38,39	325	975		
T14	2Q	26	25	27		27			4,3,14,15,38,39				
T15	3Q	15	16	14		14			4,3,27,26,38,39				
T16	4Q	39	40	38		38			4,3,27,26,14,15				

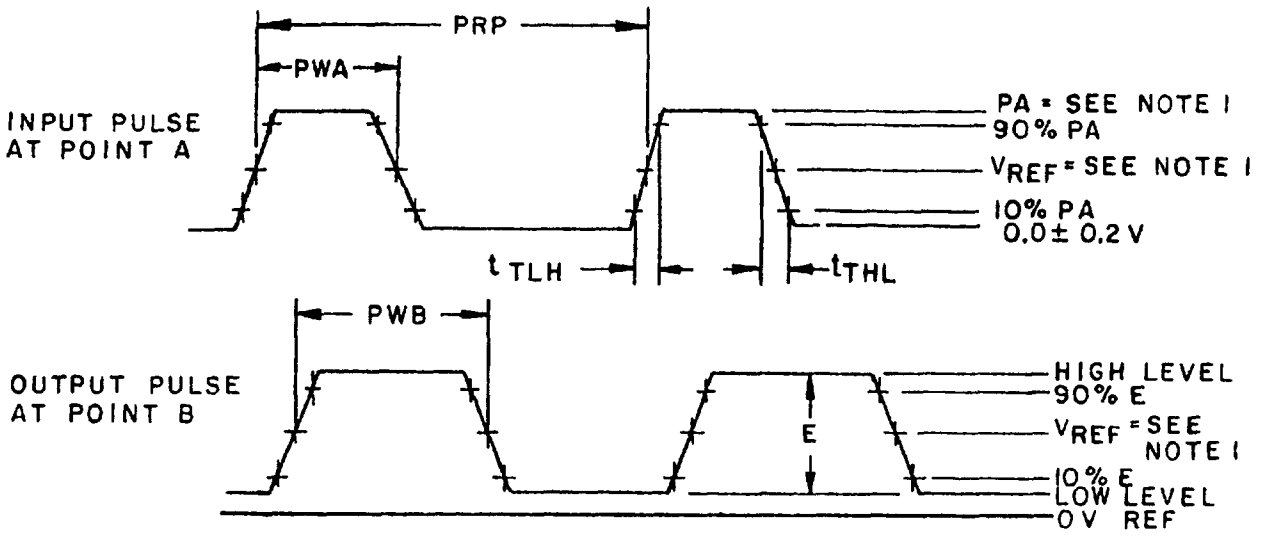
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NOTES:
1. R1 = 51.1 Ω ±5%, 1/2 W
2. R2 = 1.5 kΩ ±1%, 1/4 W
3. R3 = 100 kΩ ±1%, 1/8 W.
4. C1 = 50 ±5 pF (including probe and parasitic capacitance).
5. C2 = 1000 pF ±5%.
6. C3 = 0.01 µF ±5%.
7. Set item P07 as follows:
   5.0 V ±1% for test conditions R1-R16.
   10.0 V ±1% for test conditions S1-S16.
   15.0 V ±1% for test conditions T1-T16.

FIGURE 16 Output pulse width check test circuit for tables XVIII, XIX and XX.

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NOTES

1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PA = 5.0 ±0.1 V for test conditions R1-R16.
   =10.0 ±0.2 V for test conditions S1-S16.
   =15.0 ±0.3 V for test conditions T1-T16.
   V_{REF} = 2.5 ±0.1 V for test conditions R1-R16.
   = 5.0 ±0.2 V for test conditions S1-S16.
   = 7.5 ±0.3 V for test conditions T1-T16.
   PWA = 1.0 ±0.1 µs for test conditions R1-R8, S1-S8, T1-T8.
   = 15.0 ±2 µs for test conditions R9-R12, S9-S12.
   = 20.0 µs ±10% for test conditions T9-T12.
   = 150 µs ±10% for test conditions R13-R16, S13-S16.
   = 200 µs ±10% for test conditions T13-T16.
   PRP = 12.0 µs ±10% for test conditions R1-R8.
   = 15.0 µs ±10% for test conditions S1-S8, T1-T8.
   = 60.0 ±6 µs for test conditions R9-R12.
   = 90.0 µs ±10% for test conditions S9-S12.
   = 120 µs ±10% for test conditions T9-T12.
   = 600 µs ±10% for test conditions R13-R16.
   = 1200 µs ±10% for test conditions S13-S16.
   = 1500 µs ±10% for test conditions T13-T16.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions: R1-R16.
   ≥ 9.5 V for test conditions. S1-S16.
   ≥ 13.5 V for test conditions: T1-T16.
   Low level ≤ 0.4 V for test conditions. R1-T16.
   ≤ 0.5 V for test conditions: S1-S16.
   ≤ 1.5 V for test conditions T1-T16.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probe Y01 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground (V_{SS})

FIGURE 17. Output pulse width check test circuit waveforms for tables XVIII, XIX, and XX.

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TABLE XXI. Transition time (t_{TLH}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
U1	1Q Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	3	2	26,15,39,27,14,38	4	4	35	350	ns	18,19
U2	$1\overline{Q}$	3	21	26,15,39,27,14,38	4	4				18,20
U3	2Q	26	25	4,3,14,15,38,39	27	27				18,19
U4	$2\overline{Q}$	26	5	4,3,14,15,38,39	27	27				18,20
U5	3Q	15	16	4,3,27,26,38,39	14	14				18,19
U6	$3\overline{Q}$	15	37	4,3,27,26,38,39	14	14				18,20
U7	4Q	39	40	4,3,27,26,14,15	38	38				18,19
U8	$4\overline{Q}$	39	20	4,3,27,26,14,15	38	38				18,20
U9	1Q Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	3	2	26,15,39,27,14,38	4	4	83	830		18,19
U10	$1\overline{Q}$	3	21	26,15,39,27,14,38	4	4				18,20
U11	2Q	26	25	4,3,14,15,38,39	27	27				18,19
U12	$2\overline{Q}$	26	5	4,3,14,15,38,39	27	27				18,20
U13	3Q	15	16	4,3,27,26,38,39	14	14				18,19
U14	$3\overline{Q}$	15	37	4,3,27,26,38,39	14	14				18,20
U15	4Q	39	40	4,3,27,26,14,15	38	38				18,19
U16	$4\overline{Q}$	39	20	4,3,27,26,14,15	38	38				18,20

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TABLE XXII. Transition time (t_{THL}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	1Q Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	3	2	26,15,39,27,14,38	4	4	26	260	ns	18,19
V2	1$\overline{Q}$	3	21	26,15,39,27,14,38	4	4				18,20
V3	2Q	26	25	4,3,14,15,38,39	27	27				18,19
V4	2$\overline{Q}$	26	5	4,3,14,15,38,39	27	27				18,20
V5	3Q	15	16	4,3,27,26,38,39	14	14				18,19
V6	3$\overline{Q}$	15	37	4,3,27,26,38,39	14	14				18,20
V7	4Q	39	40	4,3,27,26,14,15	38	38				18,19
V8	4$\overline{Q}$	39	20	4,3,27,26,14,15	38	38				18,20
V9	1Q Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	3	2	26,15,39,27,14,38	4	4	74	740		18,19
V10	1$\overline{Q}$	3	21	26,15,39,27,14,38	4	4				18,20
V11	2Q	26	25	4,3,14,15,38,39	27	27				18,19
V12	2$\overline{Q}$	26	5	4,3,14,15,38,39	27	27				18,20
V13	3Q	15	16	4,3,27,26,38,39	14	14				18,19
V14	3$\overline{Q}$	15	37	4,3,27,26,38,39	14	14				18,20
V15	4Q	39	40	4,3,27,26,14,15	38	38				18,19
V16	4$\overline{Q}$	39	20	4,3,27,26,14,15	38	38				18,20

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TABLE XXIII. Transition time (t_{TLH} and t_{THL}, V_{DD} = 10 V) test.

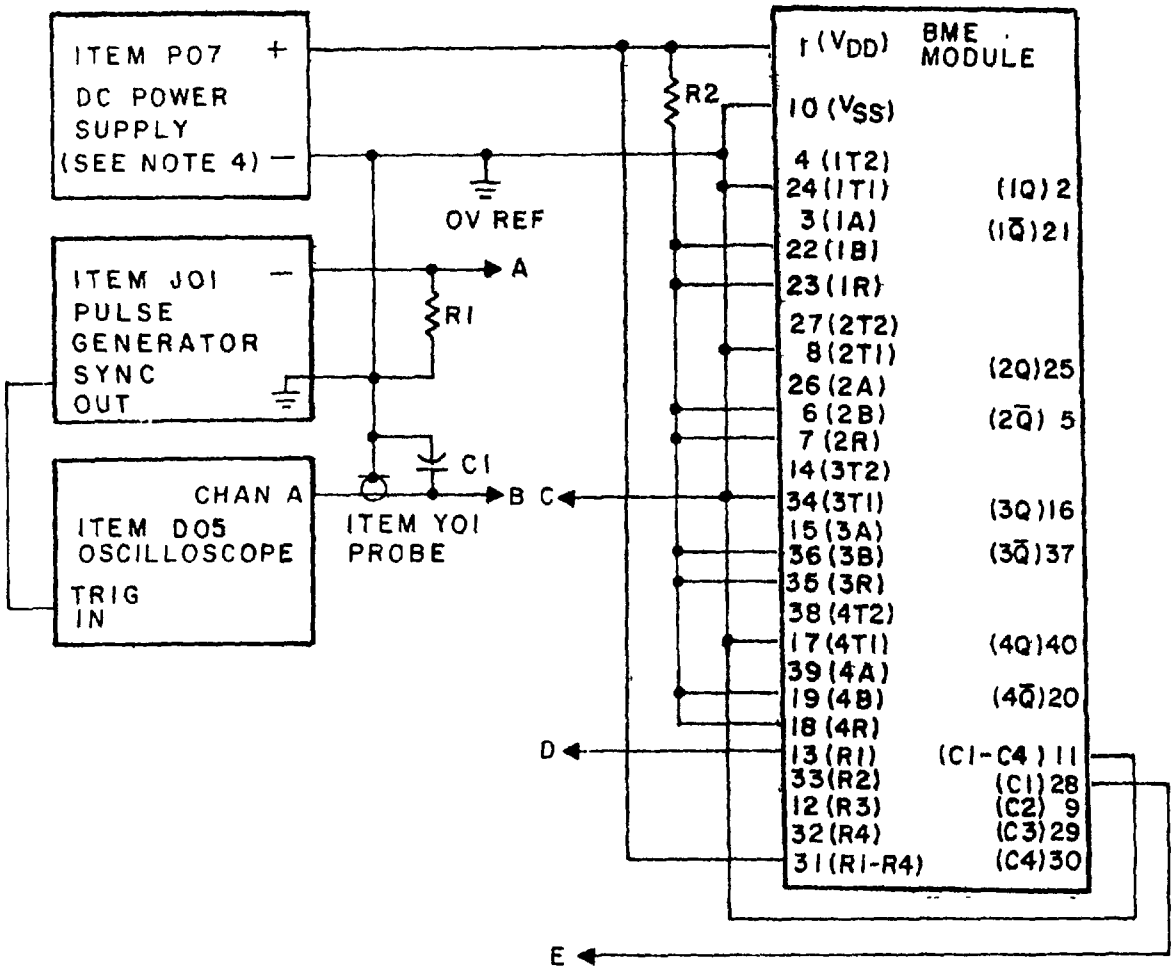
Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	1Q Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	3	2	26,15,39,27,14,38	4	4	15	150	ns	18,19
W2	1$\overline{Q}$	3	21	26,15,39,27,14,38	4	4				18,20
W3	2Q	26	25	4,3,14,15,38,39	27	27				18,19
W4	2$\overline{Q}$	26	5	4,3,14,15,38,39	27	27				18,20
W5	3Q	15	16	4,3,27,26,38,39	14	14				18,19
W6	3$\overline{Q}$	15	37	4,3,27,26,38,39	14	14				18,20
W7	4Q	39	40	4,3,27,26,14,15	38	38				18,19
W8	4$\overline{Q}$	39	20	4,3,27,26,14,15	38	38				18,20
W9	1Q Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	3	2	26,15,39,27,14,38	4	4	13	130		18,19
W10	1$\overline{Q}$	3	21	26,15,39,27,14,38	4	4				18,20
W11	2Q	26	25	4,3,14,15,38,39	27	27				18,19
W12	2$\overline{Q}$	26	5	4,3,14,15,38,39	27	27				18,20
W13	3Q	15	16	4,3,27,26,38,39	14	14				18,19
W14	3$\overline{Q}$	15	37	4,3,27,26,38,39	14	14				18,20
W15	4Q	39	40	4,3,27,26,14,15	38	38				18,19
W16	4$\overline{Q}$	39	20	4,3,27,26,14,15	38	38				18,20

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TABLE XXIV. Transition time (t_{TLH} and t_{THL}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X1	1Q Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	3	2	26,15,39,27,14,38	4	4	11	110	ns	18,19
X2	1$\overline{Q}$	3	21	26,15,39,27,14,38	4	4				18,20
X3	2Q	26	25	4,3,14,15,38,39	27	27				18,19
X4	2$\overline{Q}$	26	5	4,3,14,15,38,39	27	27				18,20
X5	3Q	15	16	4,3,27,26,38,39	14	14				18,19
X6	3$\overline{Q}$	15	37	4,3,27,26,38,39	14	14				18,20
X7	4Q	39	40	4,3,27,26,14,15	38	38				18,19
X8	4$\overline{Q}$	39	20	4,3,27,26,14,15	38	38				18,20
X9	1Q Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	3	2	26,15,39,27,14,38	4	4	10	100		18,19
X10	1$\overline{Q}$	3	21	26,15,39,27,14,38	4	4				18,20
X11	2Q	26	25	4,3,14,15,38,39	27	27				18,19
X12	2$\overline{Q}$	26	5	4,3,14,15,38,39	27	27				18,20
X13	3Q	15	16	4,3,27,26,38,39	14	14				18,19
X14	3$\overline{Q}$	15	37	4,3,27,26,38,39	14	14				18,20
X15	4Q	39	40	4,3,27,26,14,15	38	38				18,19
X16	4$\overline{Q}$	39	20	4,3,27,26,14,15	38	38				18,20

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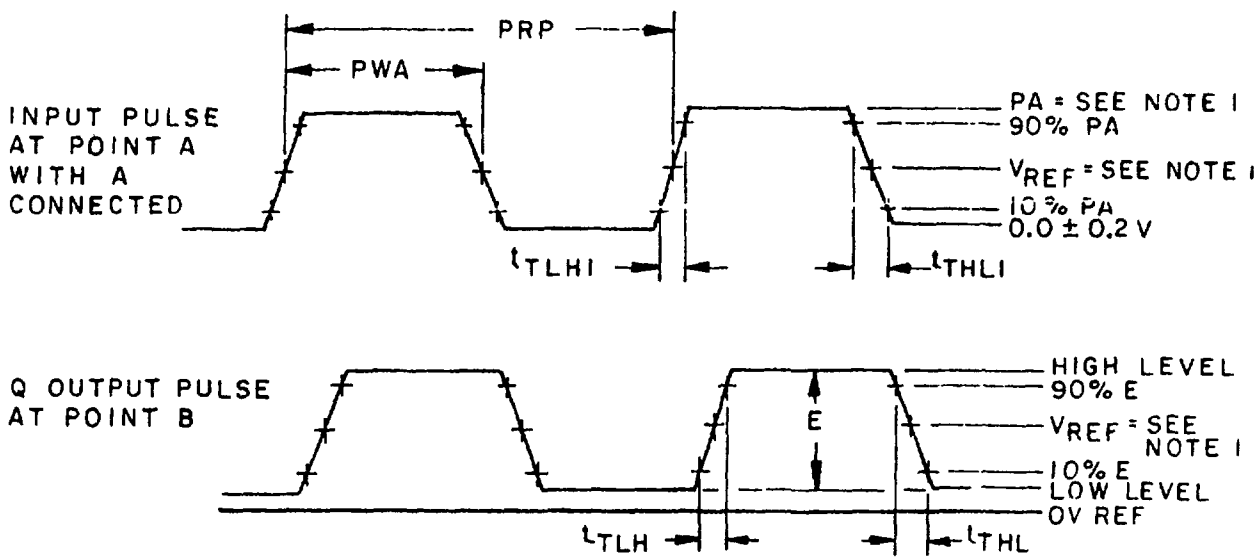


NOTES

1. R1 = 51.1 Ω ±5%, 1/2 W (V_{DD} = 5 V), 2 W (V_{DD} = 10 V), 5 W (V_{DD} = 15 V).
2. R2 = 1.5 kΩ ±1%, 1/4 W.
3. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions U1-U8, V1-V8, W1-W16, and X1-X16.
   = 150 ±15 pF (including probe and parasitic capacitance) for test conditions U9-U16 and V9-V16.
4. Set item P07 as follows.
   5.0 V ±1% for test conditions U1-U16 and V1-V16.
   10.0 V ±1% for test conditions W1-W16
   15.0 V ±1% for test conditions X1-X16.

FIGURE 18. Transition time (t_{TLH} and t_{THL}) test circuit for tables XXI through XXIV.

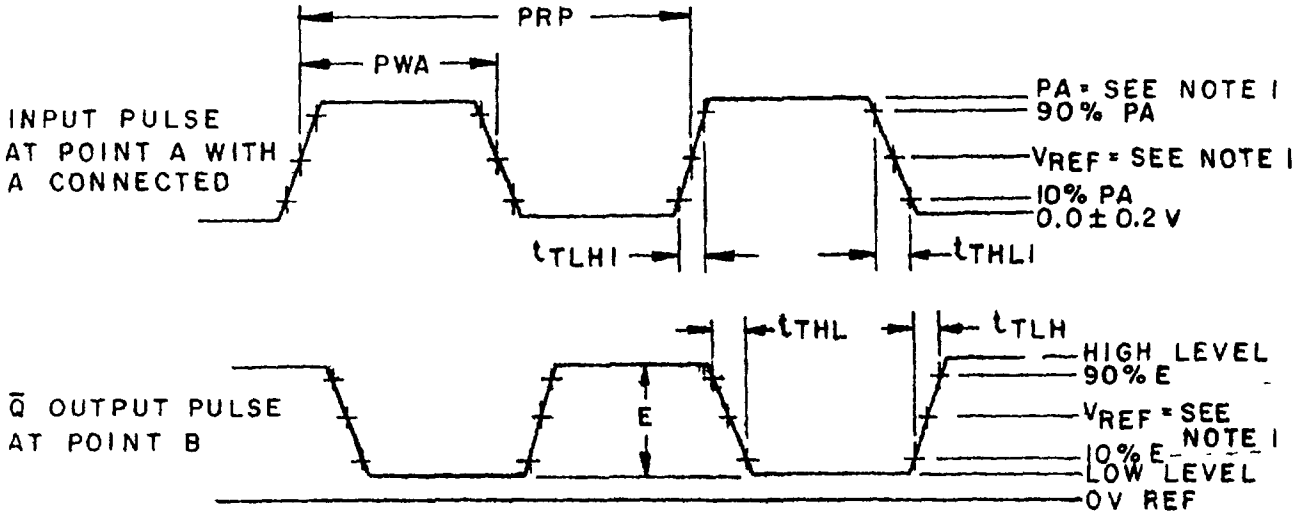
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PWA = 1.0 ±0.1 µs.
   PRP = 5.0 ±0 5 µs.
   PA = 5.0 ±0.1 V for test conditions U1-U16 and V1-V16.
   =10 0 ±0.1 V for test conditions W1-W16.
   =15 0 ±0.2 V for test conditions X1-X16.
   V_{REF} = 2.5 ±0.1 V for test conditions U1-U16 and V1-V16.
   = 5.0 ±0.1 V for test conditions W1-W16.
   = 7.5 ±0.2 V for test conditions X1-X16.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions: U1-U16 and V1-V16.
   ≥ 9.5 V for test conditions: W1-W16.
   ≥ 13.5 V for test conditions: X1-X16.
   Low level ≤ 0.4 V for test conditions: U1-U16 and V1-V16.
   ≤ 0.5 V for test conditions: W1-W16.
   ≤ 1.5 V for test conditions: X1-X16.
3. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probe Y01 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground (V_{SS})

FIGURE 19. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XXI through XXIV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns
   PWA = 1.0 ±0.1 µs.
   PRP = 5.0 ±0.5 µs.
   PA = 5.0 ±0.1 V for test conditions U1-U16 and V1-V16.
   =10 0 ±0.1 V for test conditions W1-W16.
   =15.0 ±0.2 V for test conditions X1-X16.
   V_{REF} = 2.5 ±0.1 V for test conditions U1-U16 and V1-V16.
   = 5.0 ±0.1 V for test conditions W1-W16.
   = 7.5 ±0.2 V for test conditions X1-X16.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions: U1-U16 and V1-V16
   ≥ 9.5 V for test conditions: W1-W16.
   ≥ 13.5 V for test conditions: X1-X16.
   Low level ≤ 0.4 V for test conditions: U1-U16 and V1-V16.
   ≤ 0.5 V for test conditions. W1-W16.
   ≤ 1.5 V for test conditions: X1-X16.
3. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probe Y01 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground (V_{SS})

FIGURE 20. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XXI through XXIV.

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TABLE XXV. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y1	1A to 1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	66	660	ns	21,22
Y2	1B to 1Q	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
Y3	2A to 2Q	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
Y4	2B to 2Q	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
Y5	3A to 3Q	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
Y6	3B to 3Q	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
Y7	4A to 4Q	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
Y8	4B to 4Q	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23
Y9	1A to 1Q	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	110	1100		21,22
Y10	1B to 1Q	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
Y11	2A to 2Q	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
Y12	2B to 2Q	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23

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TABLE XXV. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y13	3A to 3Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18	110	1100	ns	21,22
Y14	3B to 3Q	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
Y15	4A to 4Q	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
Y16	4B to 4Q	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23
Y17	1A to 1Q Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	86	860		21,22
Y18	1B to 1Q	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
Y19	2A to 2Q	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
Y20	2B to 2Q	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
Y21	3A to 3Q	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
Y22	3B to 3Q	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
Y23	4A to 4Q	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
Y24	4B to 4Q	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23

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TABLE XXV. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y25	1A to 1Q Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	130	1300	ns	21,22
Y26	1B to 1Q	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
Y27	2A to 2Q	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
Y28	2B to 2Q	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
Y29	3A to 3Q	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
Y30	3B to 3Q	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
Y31	4A to 4Q	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
Y32	4B to 4Q	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23

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TABLE XXVI. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z1	1A to 1$\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	66	660	ns	21,22
Z2	1B to 1$\overline{Q}$	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
Z3	2A to 2$\overline{Q}$	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
Z4	2B to 2$\overline{Q}$	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
Z5	3A to 3$\overline{Q}$	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
Z6	3B to 3$\overline{Q}$	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
Z7	4A to 4$\overline{Q}$	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
Z8	4B to 4$\overline{Q}$	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23
Z9	1A to 1$\overline{Q}$	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	110	1100		21,22
Z10	1B to 1$\overline{Q}$	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
Z11	2A to 2$\overline{Q}$	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
Z12	2B to 2$\overline{Q}$	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23

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TABLE XXVI. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z13	3A to 3$\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18	110	1100	ns	21,22
Z14	3B to 3$\overline{Q}$	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
Z15	4A to 4$\overline{Q}$	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
Z16	4B to 4$\overline{Q}$	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23
Z17	1A to 1$\overline{Q}$ Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	86	860		21,22
Z18	1B to 1$\overline{Q}$	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
Z19	2A to 2$\overline{Q}$	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
Z20	2B to 2$\overline{Q}$	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
Z21	3A to 3$\overline{Q}$	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
Z22	3B to 3$\overline{Q}$	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
Z23	4A to 4$\overline{Q}$	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
Z24	4B to 4$\overline{Q}$	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23

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TABLE XXVI. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z25	1A to 1$\overline{Q}$ Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	130	1300	ns	21,22
Z26	1B to 1$\overline{Q}$	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
Z27	2A to 2$\overline{Q}$	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
Z28	2B to 2$\overline{Q}$	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
Z29	3A to 3$\overline{Q}$	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
Z30	3B to 3$\overline{Q}$	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
Z31	4A to 4$\overline{Q}$	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
Z32	4B to 4$\overline{Q}$	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23

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TABLE XXVII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA1	1A to 1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	3	2	22,23,27,6,14,36, 38,19	4	26,7,15,35,39,18	23	230	ns	21,22
AA2	1B to 1Q	22	2	23,27,6,14,36,38, 19	4	3,26,7,15,35,39,18				21,23
AA3	2A to 2Q	26	25	6,7,4,22,14,36, 38,19	27	3,23,15,35,39,18				21,22
AA4	2B to 2Q	6	25	7,4,22,14,36,38, 19	27	26,3,15,35,39,18, 23				21,23
AA5	3A to 3Q	15	16	35,36,4,22,27,6, 38,19	14	3,23,26,7,39,18				21,22
AA6	3B to 3Q	36	16	35,4,22,27,6,38, 19	14	15,3,23,26,7,39, 18				21,23
AA7	4A to 4Q	39	40	18,19,4,22,27,6, 14,36	38	3,23,26,7,15,35				21,22
AA8	4B to 4Q	19	40	18,4,22,27,6,14, 36	38	39,3,23,26,7,15, 35				21,23
AA9	1A to 1Q	3	2	22,23,27,6,14,36, 38,19	4	26,7,15,35,39,18	45	450		21,22
AA10	1B to 1Q	22	2	23,27,6,14,36,38, 19	4	3,26,7,15,35,39,18				21,23
AA11	2A to 2Q	26	25	6,7,4,22,14,36, 38,19	27	3,23,15,35,39,18				21,22
AA12	2B to 2Q	6	25	7,4,22,14,36,38,	27	26,3,15,35,39,18,				21,23
AA13	3A to 3Q	15	16	35,36,4,22,27,6, 38,19	14	3,23,26,7,39,18				21,22
AA14	3B to 3Q	36	16	35,4,22,27,6,38, 19	14	15,3,23,26,7,39, 18				21,23
AA15	4A to 4Q	39	40	18,19,4,22,27,6, 14,36	38	3,23,26,7,15,35				21,22
AA16	4B to 4Q	19	40	18,4,22,27,6,14, 36	38	39,3,23,26,7,15, 35				21,23

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TABLE XXVIII. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB1	1A to 1$\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	23	230	ns	21,22
BB2	1B to 1$\overline{Q}$	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
BB3	2A to 2$\overline{Q}$	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
BB4	2B to 2$\overline{Q}$	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
BB5	3A to 3$\overline{Q}$	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
BB6	3B to 3$\overline{Q}$	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
BB7	4A to 4$\overline{Q}$	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
BB8	4B to 4$\overline{Q}$	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23
BB9	1A to 1$\overline{Q}$	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	45	450		21,22
BB10	1B to 1Q	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
BB11	2A to 2$\overline{Q}$	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
BB12	2B to 2$\overline{Q}$	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
BB13	3A to 3$\overline{Q}$	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
BB14	3B to 3$\overline{Q}$	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
BB15	4A to 4$\overline{Q}$	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
BB16	4B to 4$\overline{Q}$	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23

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TABLE XXIX. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test.

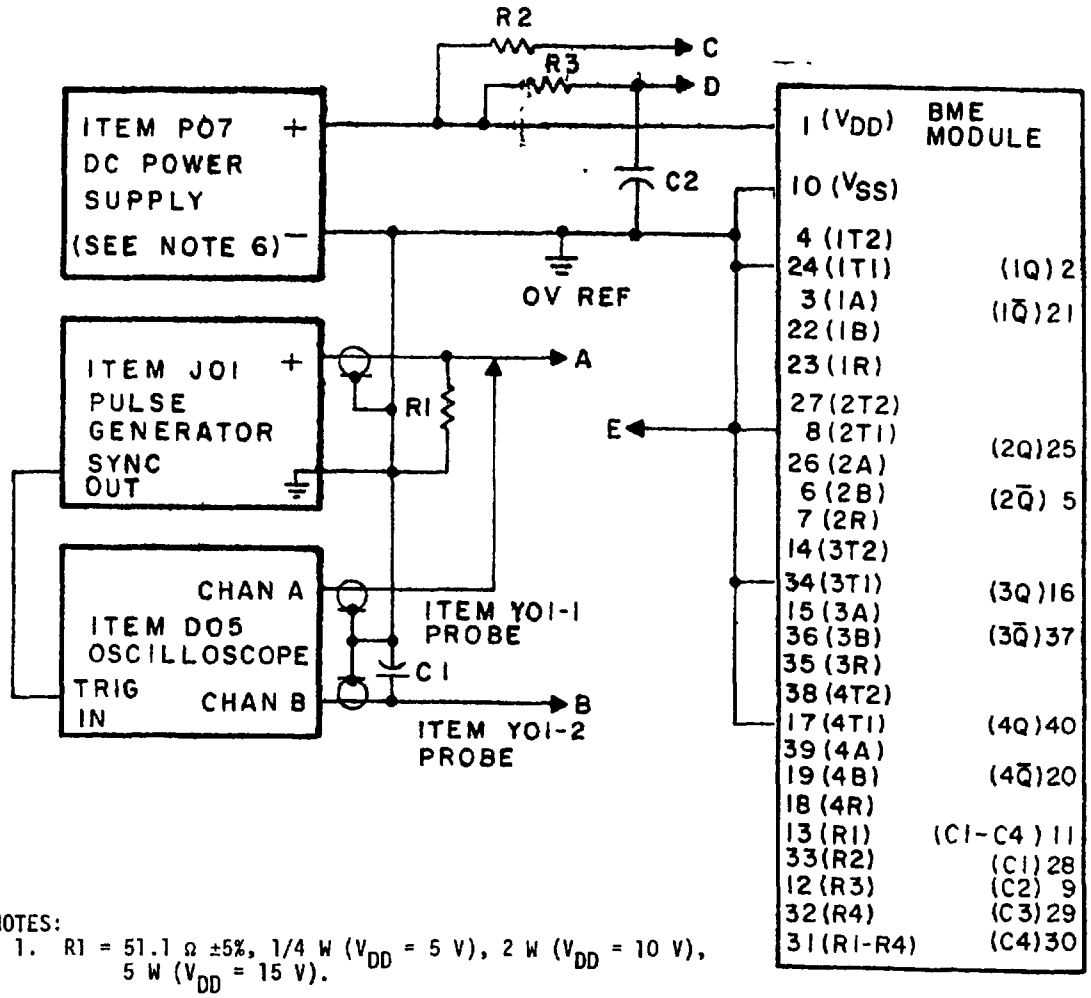
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
CC1	1A to 1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	17	175	ns	21,22
CC2	1B to 1Q	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
CC3	2A to 2Q	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
CC4	2B to 2Q	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,23				21,23
CC5	3A to 3Q	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
CC6	3B to 3Q	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
CC7	4A to 4Q	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
CC8	4B to 4Q	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23
CC9	1A to 1Q	3	2	22,23,27,6,14,36,38,19	4	26,7,15,35,39,18	32	325		21,22
CC10	1B to 1Q	22	2	23,27,6,14,36,38,19	4	3,26,7,15,35,39,18				21,23
CC11	2A to 2Q	26	25	6,7,4,22,14,36,38,19	27	3,23,15,35,39,18				21,22
CC12	2B to 2Q	6	25	7,4,22,14,36,38,19	27	26,3,15,35,39,18,				21,23
CC13	3A to 3Q	15	16	35,36,4,22,27,6,38,19	14	3,23,26,7,39,18				21,22
CC14	3B to 3Q	36	16	35,4,22,27,6,38,19	14	15,3,23,26,7,39,18				21,23
CC15	4A to 4Q	39	40	18,19,4,22,27,6,14,36	38	3,23,26,7,15,35				21,22
CC16	4B to 4Q	19	40	18,4,22,27,6,14,36	38	39,3,23,26,7,15,35				21,23

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TABLE XXX. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
DD1	1A to 1$\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	3	2	22,23,27,6,14,36, 38,19	4	26,7,15,35,39,18	17	175	ns	21,22
DD2	1B to 1$\overline{Q}$	22	2	23,27,6,14,36,38, 19	4	3,26,7,15,35,39,18				21,23
DD3	2A to 2$\overline{Q}$	26	25	6,7,4,22,14,36, 38,19	27	3,23,15,35,39,18				21,22
DD4	2B to 2$\overline{Q}$	6	25	7,4,22,14,36,38, 19	27	26,3,15,35,39,18, 23				21,23
DD5	3A to 3$\overline{Q}$	15	16	35,36,4,22,27,6, 38,19	14	3,23,26,7,39,18				21,22
DD6	3B to 3$\overline{Q}$	36	16	35,4,22,27,6,38, 38,19	14	15,3,23,26,7,39, 18				21,23
DD7	4A to 4$\overline{Q}$	39	40	18,19,4,22,27,6, 14,36	38	3,23,26,7,15,35				21,22
DD8	4B to 4$\overline{Q}$	19	40	18,4,22,27,6,14, 36	38	39,3,23,26,7,15, 35				21,23
DD9	1A to 1$\overline{Q}$	3	2	22,23,27,6,14,36, 38,19	4	26,7,15,35,39,18	32	325		21,22
DD10	1B to 1$\overline{Q}$	22	2	23,27,6,14,36,38, 19	4	3,26,7,15,35,39,18				21,23
DD11	2A to 2$\overline{Q}$	26	25	6,7,4,22,14,36, 38,19	27	3,23,15,35,39,18				21,22
DD12	2B to 2$\overline{Q}$	6	25	7,4,22,14,36,38, 19	27	26,3,15,35,39,18, 23				21,23
DD13	3A to 3$\overline{Q}$	15	16	35,36,4,22,27,6, 38,19	14	3,23,26,7,39,18				21,22
DD14	3B to 3$\overline{Q}$	36	16	35,4,22,27,6,38, 19	14	15,3,23,26,7,39, 18				21,23
DD15	4A to 4$\overline{Q}$	39	40	18,19,4,22,27,6, 14,36	38	3,23,26,7,15,35				21,22
DD16	4B to 4$\overline{Q}$	19	40	18,4,22,27,6,14, 36	38	39,3,23,26,7,15, 35				21,23

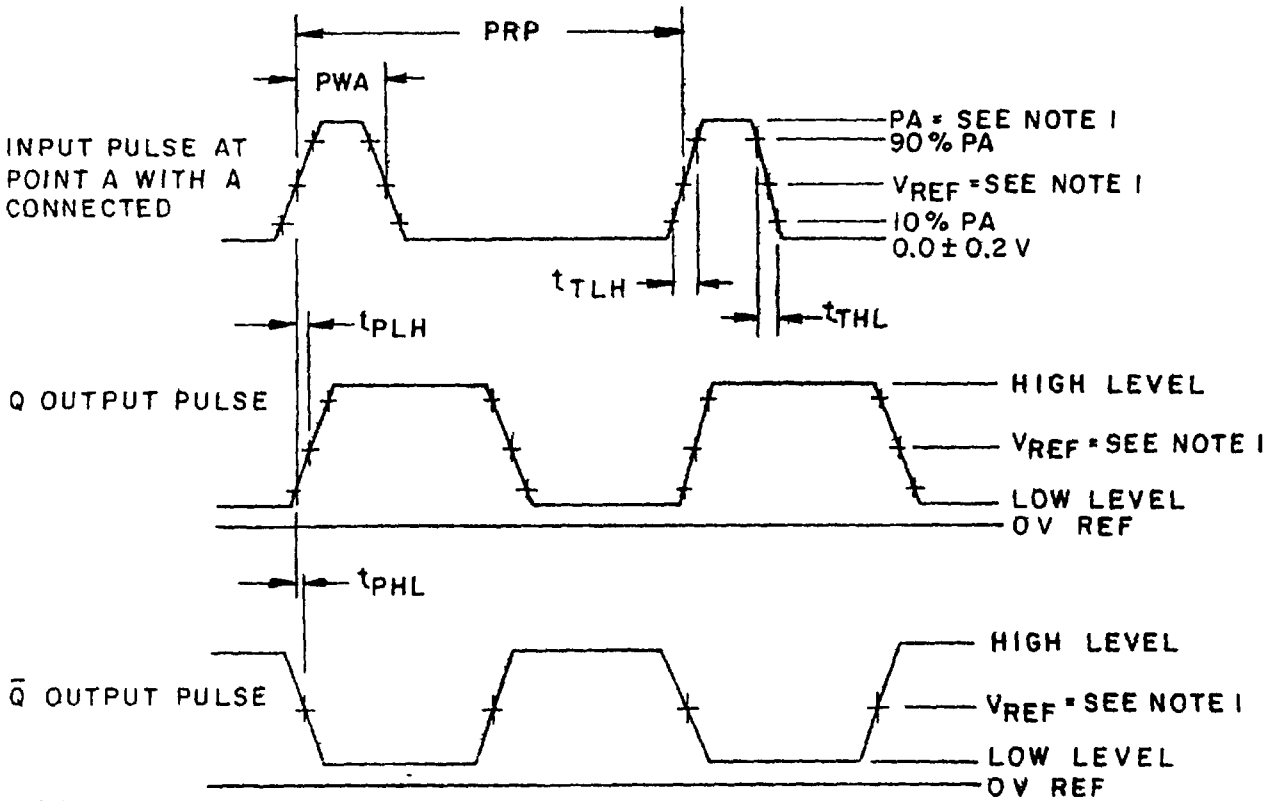
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NOTES:
1. R1 = 51.1 Ω ±5%, 1/4 W (V_{DD} = 5 V), 2 W (V_{DD} = 10 V), 5 W (V_{DD} = 15 V).
2. R2 = 1.5 kΩ ±1%, 1/4 W.
3. R3 = 10.0 kΩ ±1%, 1/8 W.
4. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions Y1-Y16, Z1-Z16, AA1-AA16, BB1-BB16, CC1-CC16 and DD1-DD16.
   = 150 ±15 pF (including probe and parasitic capacitance) for test conditions Y17-Y32 and Z17-Z32.
5. C2 = 220 pF ±5% (including parasitic capacitance) for test conditions Y1-Y8, Y17-Y24, Z1-Z8, Z17-Z24, AA1-AA8, BB1-BB8, CC1-CC8 and DD1-DD8.
   = 1000 pF ±5% (including parasitic capacitance) for test conditions Y9-Y16, Y25-Y32, Z9-Z16, Z25-Z32, AA9-AA16, BB9-BB16, CC9-CC16 and DD9-DD16.
6. Set item P07 as follows.
   5.0 V ±1% for test conditions Y1-Y32 and Z1-Z32
   10.0 V ±1% for test conditions AA1-AA16 and BB1-BB16.
   15.0 V ±1% for test conditions CC1-CC16 and DD1-DD16.

FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXV through XXX.

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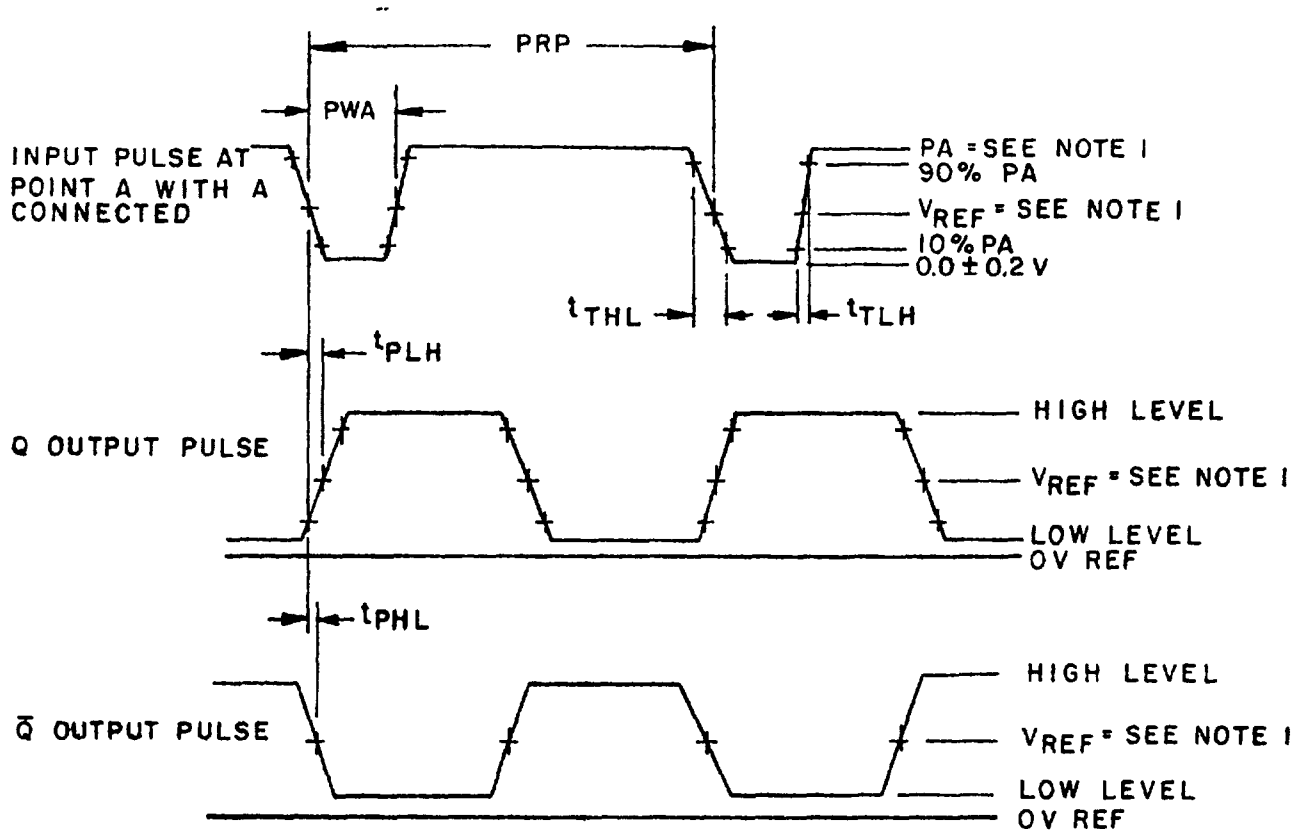


NOTES:

1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PRP = 12.0 µs ±10%.
   PWA = 150 +15, -0 ns for test conditions Y1, Y3, Y5, Y7, Y9, Y11, Y13, Y15, Y17, Y19, Y21, Y23, Y25, Y27, Y29, Y31, Z1, Z3, Z5, Z7, Z9, Z11, Z13, Z15, Z17, Z19, Z21, Z23, Z25, Z27, Z29, Z31.
   = 75 +7, -0 ns for test conditions AA1, AA3, AA5, AA7, AA9, AA11, AA13, AA15, BB1, BB3, BB5, BB7, BB9, BB11, BB13, BB15.
   = 55 +5, -0 ns for test conditions CC1, CC3, CC5, CC7, CC9, CC11, CC13, CC15, DD1, DD3, DD5, DD7, DD9, DD11, DD13, DD15.
   PA = 5.0 ±0.1 V for test conditions Y1-Y32 and Z1-Z32.
   =10.0 ±0.2 V for test conditions AA1-AA16 and BB1-BB16.
   =15.0 ±0.2 V for test conditions CC1-CC16 and DD1-DD16.
   V_{REF} = 2.5 ±0.1 V for test conditions Y1-Y32 and Z1-Z32.
   = 5.0 ±0.1 V for test conditions AA1-AA16 and BB1-BB16.
   = 7.5 ±0.1 V for test conditions CC1-CC16 and DD1-DD16.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows.
   High level ≥ 4.6 V for test conditions. Y1-Y32 and Z1-Z32.
   ≥ 9.5 V for test conditions. AA1-AA16 and BB1-BB16.
   ≥ 13.5 V for test conditions: CC1-CC16 and DD1-DD16.
   Low level ≤ 0.4 V for test conditions: Y1-Y32 and Z1-Z32.
   ≤ 0.5 V for test conditions AA1-AA16 and BB1-BB16.
   ≤ 1.5 V for test conditions: CC1-CC16 and DD1-DD16.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes YO1-1 and YO1-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm)
4. All voltages are referenced to ground (V_{SS})

FIGURE 22. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXV through XXX

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NOTES.

1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRP = 12.0 µs ±10%.
   PWA = 150 +15, -0 ns for test conditions Y2, Y4, Y6, Y8, Y10, Y12, Y14, Y16, Y18, Y20, Y22, Y24, Y26, Y28, Y30, Y32, Z2, Z4, Z6, Z8, Z10, Z12, Z14, Z16, Z18, Z20, Z22, Z24, Z26, Z28, Z30, Z32.
   = 75 +7, -0 ns for test conditions AA2, AA4, AA6, AA8, AA10, AA12, AA14, AA16, BB2, BB4, BB6, BB8, BB10, BB12, BB14, BB16.
   = 55 +5, -0 ns for test conditions CC2, CC4, CC6, CC8, CC10, CC12, CC14, CC16, DD2, DD4, DD6, DD8, DD10, DD12, DD14, DD16.
   PA = 5.0 ±0.1 V for test conditions Y1-Y32 and Z1-Z32.
   =10.0 ±0.2 V for test conditions AA1-AA16 and BB1-BB16.
   =15.0 ±0.2 V for test conditions CC1-CC16 and DD1-DD16.
   V_{REF} = 2.5 ±0.1 V for test conditions Y1-Y32 and Z1-Z32.
   = 5.0 ±0.1 V for test conditions AA1-AA16 and BB1-BB16.
   = 7.5 ±0.1 V for test conditions CC1-CC16 and DD1-DD16.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions: Y1-Y32 and Z1-Z32.
   ≥ 9.5 V for test conditions: AA1-AA16 and BB1-BB16.
   ≥ 13.5 V for test conditions: CC1-CC16 and DD1-DD16.
   Low level ≤ 0.4 V for test conditions: Y1-Y32 and Z1-Z32.
   ≤ 0.5 V for test conditions: AA1-AA16 and BB1-BB16.
   ≤ 1.5 V for test conditions: CC1-CC16 and DD1-DD16.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground (V_{SS}).

FIGURE 23. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXV through XXX.

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TABLE XXXI. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
EE1	1R to 1Q Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39,18	23	66	660	ns	24,25
EE2	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39,18	7				
EE3	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39,18	35				
EE4	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15,35	18				
EE5	1R to 1Q	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39,18	23	110	1100		
EE6	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39,18	7				
EE7	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39,18	35				
EE8	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15,35	18				

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TABLE XXXI. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
EE9	1R to 1Q Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	86	860	ns	24,25
EE10	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
EE11	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
EE12	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				
EE13	1R to 1Q	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	130	1300		
EE14	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
EE15	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
EE16	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				

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TABLE XXXII. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
FF1	1R to 1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	66	660	ns	24,25
FF2	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
FF3	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
FF4	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				
FF5	1R to 1Q	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	110	1100		
FF6	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
FF7	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
FF8	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				

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TABLE XXXII. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
FF9	1R to 1Q Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	86	860	ns	24,25
FF10	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
FF11	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
FF12	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				
FF13	1R to 1Q	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	130	1300		
FF14	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
FF15	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
FF16	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				

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TABLE XXXIII. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
GG1	1R to 1Q Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39,18	23	23	230	ns	24,25
GG2	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39,18	7				
GG3	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39,18	35				
GG4	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15,35	18				
GG5	1R to 1Q	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39,18	23	60	600		
GG6	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39,18	7				
GG7	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39,18	35				
GG8	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15,35	18				

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TABLE XXXIV. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
HH1	1R to 1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	23	230	ns	24,25
HH2	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
HH3	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
HH4	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				
HH5	1R to 1Q	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	60	600		
HH6	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
HH7	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
HH8	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				

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TABLE XXXV. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test.

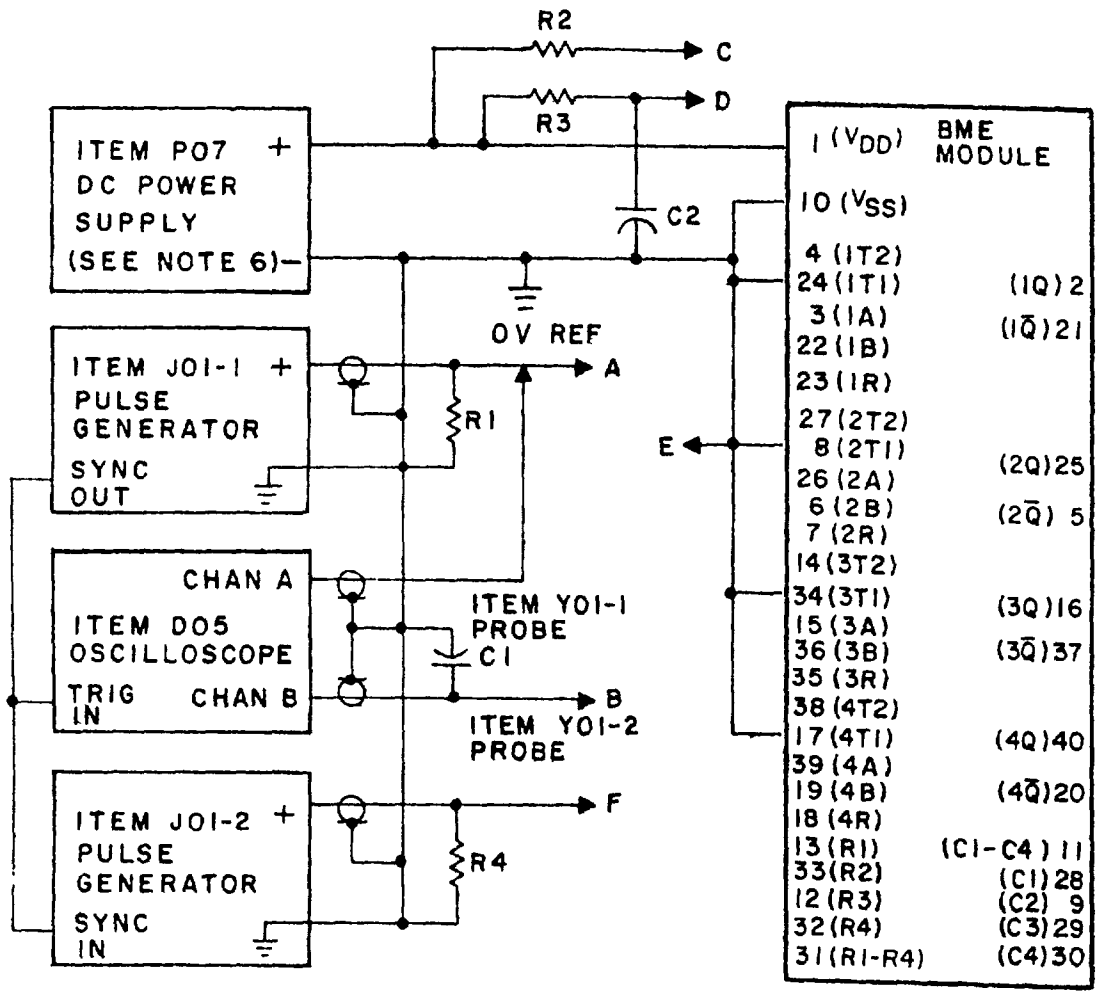
Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
JJ1	1R to 1Q Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39,18	23	17	175	ns	24,25
JJ2	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39,18	7				
JJ3	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39,18	35				
JJ4	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15,35	18				
JJ5	1R to 1Q	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39,18	23	50	500		
JJ6	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39,18	7				
JJ7	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39,18	35				
JJ8	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15,35	18				

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TABLE XXXVI. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
KK1	1R to 1Q Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	17	175	ns	24,25
KK2	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
KK3	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
KK4	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				
KK5	1R to 1Q	3	2	22,27,6,14,36,38,19	4	26,7,15,35,39, 18	23	50	500		
KK6	2R to 2Q	26	25	6,4,22,14,36,38,19	27	3,23,15,35,39, 18	7				
KK7	3R to 3Q	15	16	36,4,22,27,6,38,19	14	3,23,26,7,39, 18	35				
KK8	4R to 4Q	39	40	4,22,27,6,14,19,36	38	3,23,26,7,15, 35	18				

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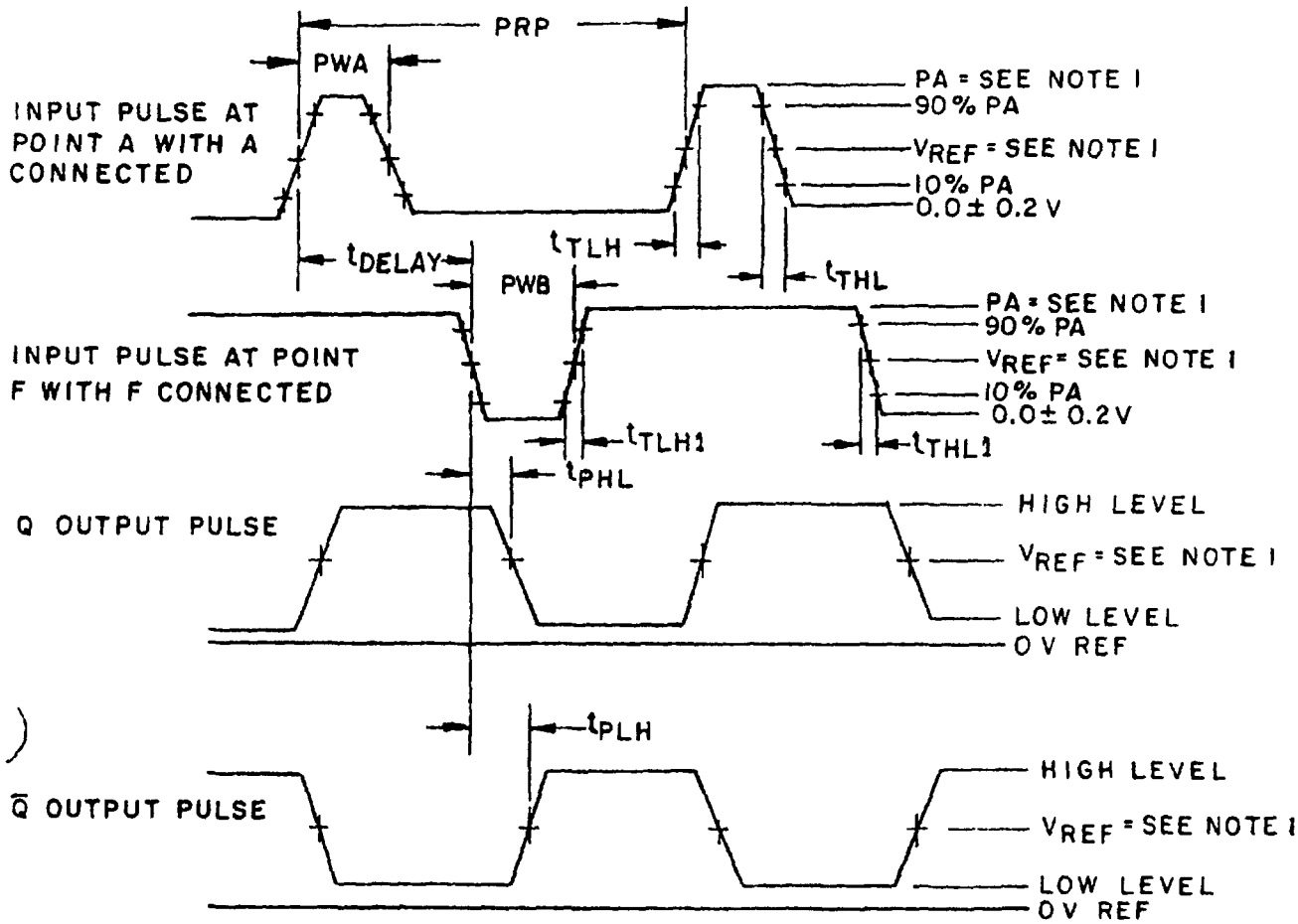


NOTES.

1. R1 = R4 = 51.1 Ω ±5%, 1/2 (V_{DD} = 5 V), 2 W (V_{DD} = 10 V), 5 W (V_{DD} = 15 V).
2. R2 = 1.5 kΩ ±1%, 1/4 W.
3. R3 = 10 kΩ ±1%, 1/8 W.
4. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions EE1-EE8, FF1-FF8, GG1-GG8, HH1-HH8, JJ1-JJ8, and KK1-KK8.
   = 150 ±15 pF (including probe and parasitic capacitance) for test conditions EE9-EE16 and FF9-FF16.
5. C2 = 220 pF ±5% (including probe and parasitic capacitance) for test conditions EE1-EE4, EE9-EE12, FF1-FF4, FF9-FF12, GG1-GG4, HH1-HH4, JJ1-JJ4 and KK1-KK4.
   = 1000 pF ±5% (including probe and parasitic capacitance) for test conditions EE5-EE8, EE13-EE16, FF5-FF8, FF13-FF16, GG5-GG8, HH5-HH8, JJ5-JJ8 and KK5-KK8.
6. Set item PO7 as follows:
   5.0 V ±1 for test conditions EE1-EE16 and FF1-FF16.
   10.0 V ±1% for test conditions GG1-GG8 and HH1-HH8
   15.0 V ±1% for test conditions JJ1-JJ8 and KK1-KK8

FIGURE 24. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XXXI through XXXVI.

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NOTES:

1. Input pulse parameters:
   $t_{TLH} = t_{TLH1} = t_{THL} = t_{THL1} = 10 \pm 2$ ns
   PRP = 12 µs ±10%.
   t_{DELAY} = 1 µs ±10%.
   PWA = PWB = 150 +15, -0 ns for test conditions EE1-EE16 and FF1-FF16
   = 75 +7, -0 ns for test conditions GG1-GG8 and HH1-HH8.
   = 55 +5, -0 ns for test conditions JJ1-JJ8 and KK1-KK8.
   PA = 5.0 V ±1% for test conditions EE1-EE16 and FF1-FF16.
   = 10.0 V ±1% for test conditions GG1-GG8 and HH1-HH8
   = 15.0 V ±1% for test conditions JJ1-JJ8 and KK1-KK8.
   V_{REF} = 2.5 V ±1% for test conditions EE1-EE16 and FF1-FF16.
   = 5.0 V ±1% for test conditions GG1-GG8 and HH1-HH8.
   = 15.0 V ±1% for test conditions JJ1-JJ8 and KK1-KK8
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions: EE1-EE16 and FF1-FF16.
   ≥ 9.5 V for test conditions: GG1-GG8 and HH1-HH8.
   ≥ 13.5 V for test conditions: JJ1-JJ8 and KK1-KK8.
   Low level ≤ 0.4 V for test conditions: EE1-EE16 and FF1-FF16.
   ≤ 0.5 V for test conditions: GG1-GG8 and HH1-HH8.
   ≤ 1.5 V for test conditions: JJ1-JJ8 and KK1-KK8.
3. The lead length of electrical conductors between terminator resistor R1 and R3 and input pins, probes YO1-1 and YO1-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground (V_{SS}).

FIGURE 25 Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XXXI through XXXVI

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TABLE XXXVII. Output retriggerability check (V_{DD} = 5 V) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
LL1	1Q Output retriggerability check (V_{DD} = 5 V)	3	2	4	4	27,26,14,15,38, 39	1.65	6.70	µs	26,27
LL2	2Q	26	25	27	27	4,3,14,15,38,39				
LL3	3Q	15	16	14	14	4,3,27,26,38,39				
LL4	4Q	39	40	38	38	4,3,27,26,14,15				

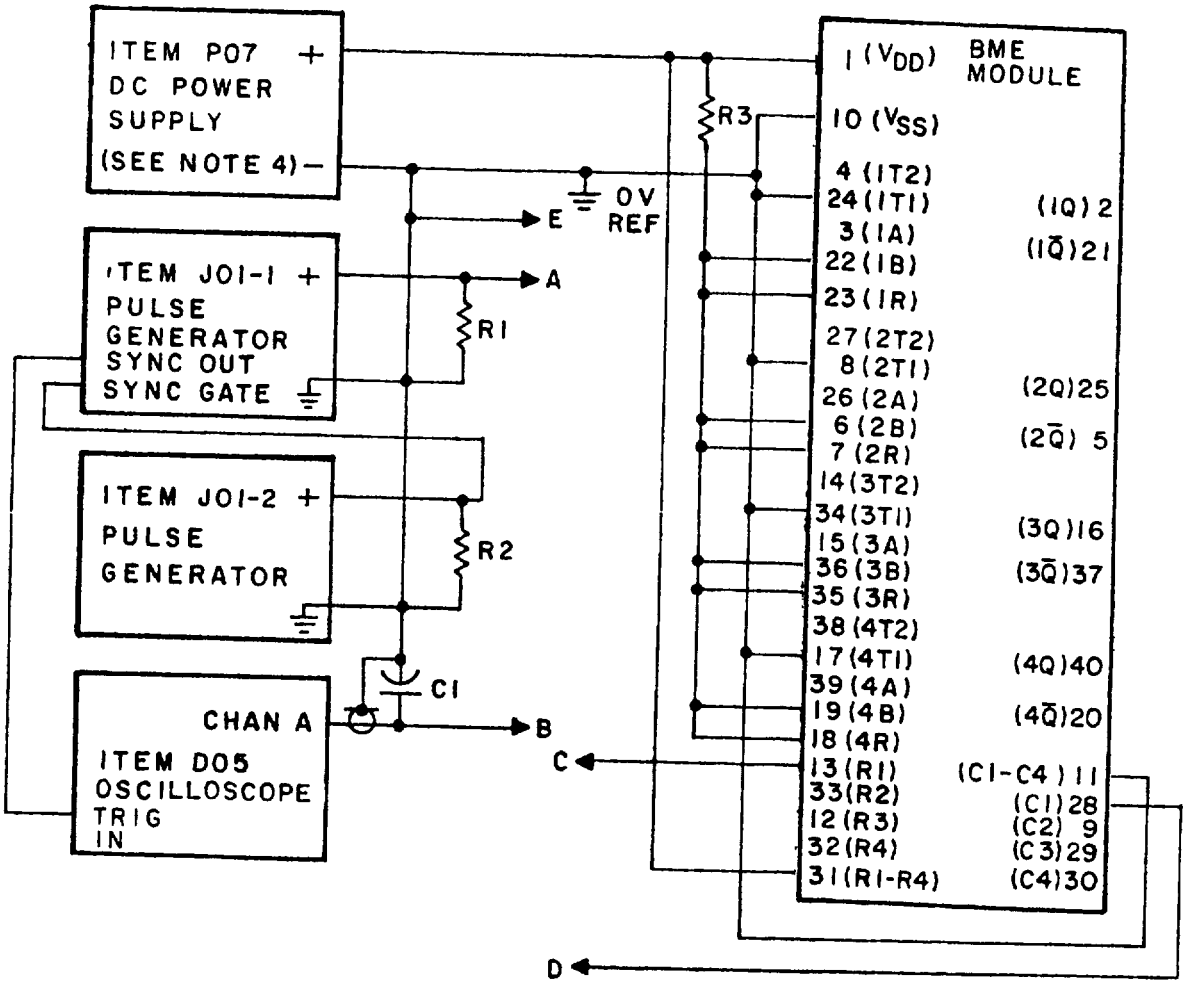
TABLE XXXVIII. Output retriggerability check (V_{DD} = 10 V) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
MM1	1Q Output retriggerability check (V_{DD} = 10 V)	3	2	4	4	27,26,14,15,38, 39	1.25	7.50	µs	26,27
MM2	2Q	26	25	27	27	4,3,14,15,38,39				
MM3	3Q	15	16	14	14	4,3,27,26,38,39				
MM4	4Q	39	40	38	38	4,3,27,26,14,15				

TABLE XXXIX. Output retriggerability check (V_{DD} = 15 V) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
NN1	1Q Output retriggerability check (V_{DD} = 15 V)	3	2	4	4	27,26,14,15,38, 39	1.25	7.50	µs	26,27
NN2	2Q	26	25	27	27	4,3,14,15,38,39				
NN3	3Q	15	16	14	14	4,3,27,26,38,39				
NN4	4Q	39	40	38	38	4,3,27,26,14,15				

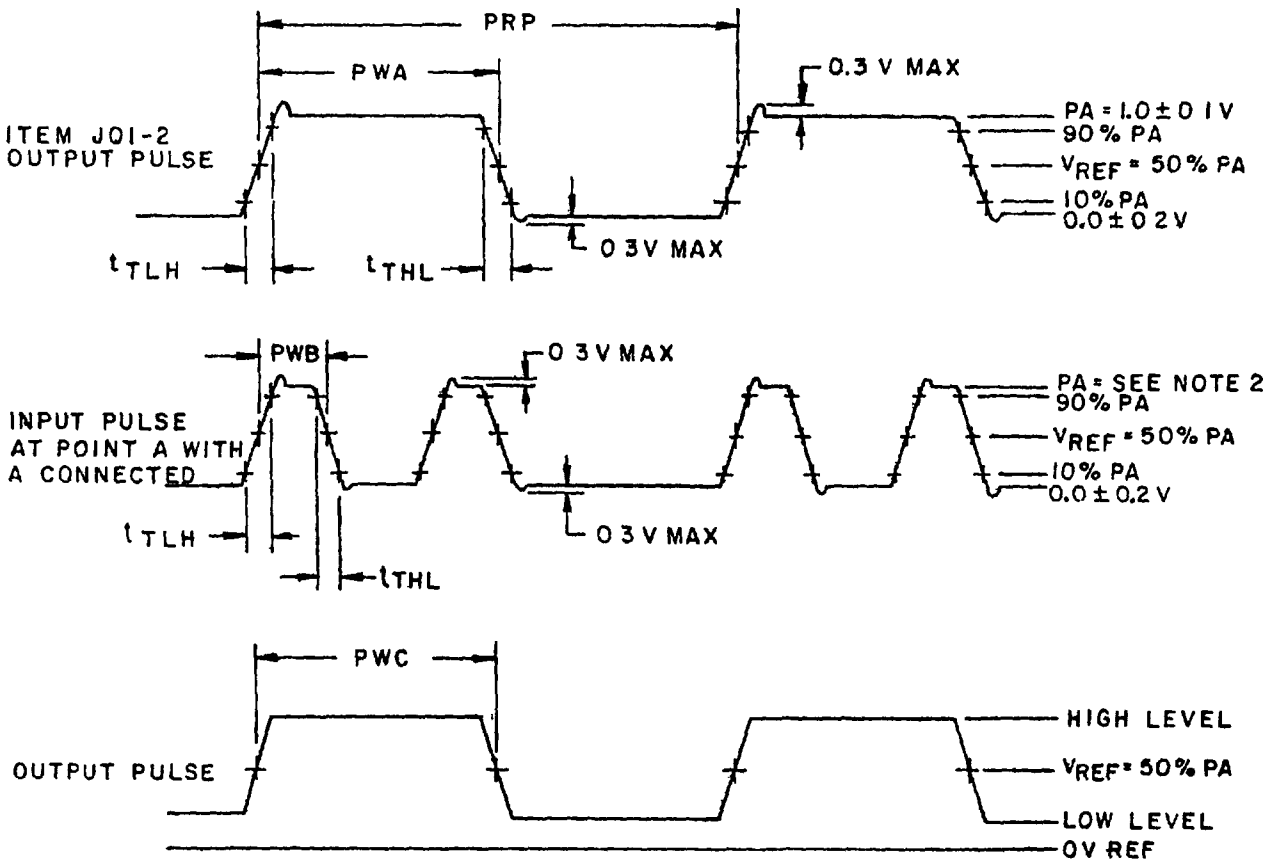
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NOTES.
1. R1 = R2 = 51.1 Ω ±5%, 1/2 W.
2. R3 = 1.5 kΩ ±1%, 1/4 W.
3. C1 = 50 ±5 pF (including probe and parasitic capacitance).
4. Set item P07 as follows:
   5.0 V ±1% for test conditions LL1-LL4.
   10.0 V ±1% for test conditions MM1-MM4.
   15.0 V ±1% for test conditions NN1-NN4.

FIGURE 26. Output retriggerability check test circuit for tables XXXVII, XXXVIII and XXXIX.

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NOTES:

1. Input pulse parameters (item J01-2).
   PWA = 1.0 µs ±100 ns.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Input pulse parameters (item J01-1)
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWB = 150 +15, -0 ns for test conditions LL1-LL4.
   = 75 +7, -0 ns for test conditions MM1-MM4
   = 55 +5, -0 ns for test conditions NN1-NN4
   PA = 5 ±0.1 V for test conditions LL1-LL4
   = 10 ±0.2 V for test conditions MM1-MM4.
   = 15 ±0.2 V for test conditions NN1-NN4
3. Before making the measurements specified, verify that the output pulse high level and low level is as follows.
   High level ≥ 4.5 V for test conditions LL1-LL4.
   ≥ 9.0 V for test conditions MM1-MM4
   ≥ 13.5 V for test conditions NN1-NN4
   Low level ≤ 0.5 V for test conditions LL1-LL4.
   ≤ 1.0 V for test conditions MM1-MM4.
   ≤ 1.5 V for test conditions NN1-NN4

FIGURE 27 Output retriggerability check test circuit waveforms for tables XXXVII, XXXVIII and XXXIX.

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TABLE XL. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
PP1	R1 Resistance	13	31	9.9	10.1	kΩ	28
PP2	R2	33					
PP3	R3	12					
PP4	R4	32					

TABLE XLI. Power filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
QQ	Power supply filter capacitance	1	Open	0.007	0.013	µF	29

TABLE XLII. Capacitance test.

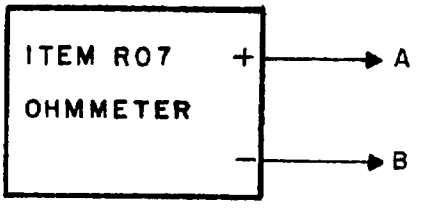
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
RR1	C1 Capacitance	Open	28	11	198	242	pF	29
RR2	C2		9					
RR3	C3		29					
RR4	C4		30					

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TABLE XLIII. Input capacitance test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
SS1	1T2 Input capacitance	Open	4	10	2	15	pF	29
SS2	1A		3					
SS3	1B		22					
SS4	1R		23					
SS5	2T2		27					
SS6	2A		26					
SS7	2B		6					
SS8	2R		7					
SS9	3T2		14					
SS10	3A		15					
SS11	3B		36					
SS12	3R		35					
SS13	4T2		38					
SS14	4A		39					
SS15	4B		19					
SS16	4R		18					

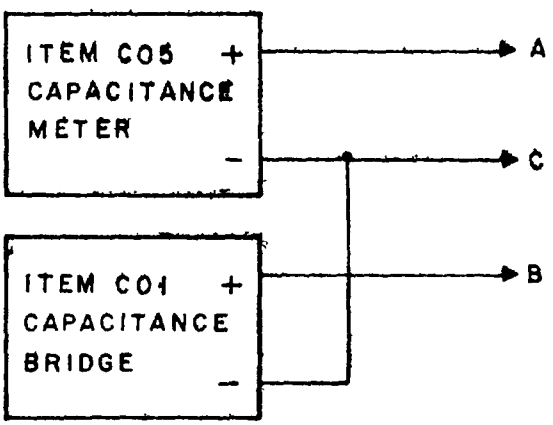
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	BME MODULE
1 (V_{DD})	
10 (V_{SS})	
4 (1T2)	
24 (1T1)	(1Q) 2
3 (1A)	
22 (1B)	(1$\bar{Q}$) 21
23 (1R)	
27 (2T2)	
8 (2T1)	
26 (2A)	(2Q) 25
6 (2B)	
7 (2R)	(2$\bar{Q}$) 5
14 (3T2)	
34 (3T1)	(3Q) 16
15 (3A)	
36 (3B)	(3$\bar{Q}$) 37
35 (3R)	
38 (4T2)	
17 (4T1)	(4Q) 40
39 (4A)	
19 (4B)	(4$\bar{Q}$) 20
18 (4R)	
13 (R1)	(C1-C4) 11
33 (R2)	(C1) 28
12 (R3)	(C2) 9
32 (R4)	(C3) 29
31 (R1-R4)	(C4) 30

FIGURE 28. Resistance test circuit for table XL.

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1 (V_{DD})	BME MODULE
10 (V_{SS})	
4 (1T2)	
24 (1T1)	(1Q) 2
3 (1A)	(1$\bar{Q}$) 21
22 (1B)	
23 (1R)	
27 (2T2)	
8 (2T1)	(2Q) 25
26 (2A)	
6 (2B)	(2$\bar{Q}$) 5
7 (2R)	
14 (3T2)	
34 (3T1)	(3Q) 16
15 (3A)	
36 (3B)	(3$\bar{Q}$) 37
35 (3R)	
38 (4T2)	
17 (4T1)	(4Q) 40
39 (4A)	
19 (4B)	(4$\bar{Q}$) 20
18 (4R)	
13 (R1)	(C1-C4) 11
33 (R2)	(C1) 28
12 (R3)	(C2) 9
32 (R4)	(C3) 29
31 (R1-R4)	(C4) 30

NOTES.
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10 or 11, or the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge C01 setup:
   Frequency = 1.0 kHz.
   V_{pk} = 0.3 V maximum

FIGURE 29. Power supply filter capacitance, capacitance (C1, C2, C3, C4) and input capacitance test circuit for tables XLI, XLII and XLIII

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